

The Incidence of Carbon Taxes in an Electrifying Economy

Sevim Dinlemez*

This version: May, 2026 (first version: March 2026)

Working paper

[[Link to most recent version](#)]

Abstract

I study how household electrification shapes the distributional welfare effects of a carbon tax, focusing on the role of revenue recycling instruments. I develop a DSGE model with a discrete household electrification decision subject to financing frictions that slow adoption among low-income households, and non-homothetic preferences for energy. Using green subsidies as a revenue-recycling instrument lowers the peak tax rate required to achieve a given climate target, as it increases the share of renewable sources in electricity production and shifts energy demand towards electricity rather than fossil fuels. The lower tax rate and lower electricity price attenuate distortions in economic growth. However, as richer households electrify earlier, recycling carbon tax revenues through green subsidies disproportionately benefits higher-income households, leading to unequal welfare effects. Lump-sum transfers provide better insurance to financially constrained households but come at the cost of larger GDP and labor income losses. As an extension, I show that front-loaded carbon taxes reach climate targets with lower peak tax rates, limiting wage and GDP losses, but can amplify short-run inequality when combined with subsidy recycling.

Keywords: Green transition, Climate Policy, Welfare, Inequality, Electrification

*University of Amsterdam, Faculty of Economics and Business, 1018 WB, Amsterdam, The Netherlands. Email: s.e.g.dinlemez@uva.nl. I am thankful for the guidance of my advisors Marcelo Pedroni and Fabian Greimel. I am grateful for the help of Christian Stoltenberg, Alessia Russo, Daniel Schmidt, Lennard Schlattmann, Taco Prins, and all participants of seminars/conferences at the University of Amsterdam, University of Vienna, University of Padua. All errors are my own.

1 Introduction

Carbon pricing is widely regarded as essential to reduce emissions and global warming, but it can also raise distributional concerns. Carbon taxes may impose relatively larger burdens on poorer households. This distributional incidence of the policy depends not only on the tax itself, but also on how carbon tax revenues are recycled. Policymakers can choose to return revenues directly to households, subsidize green technology adoption or renewable electricity production, or finance government expenditure. This decision involves a trade-off between efficiency and equity: green subsidies may lower the peak carbon tax required to curb emissions and thus reduce tax distortions, but they may also be regressive if higher-income households can more easily adapt to climate policy and reap the benefits of the subsidies. In this paper, I study how household electrification as a form of climate policy adaptation shapes this trade-off. Electrification requires large upfront investments in green technologies (e.g. electric vehicles, electric heating systems) and adoption is unequally distributed across the income distribution. Poor households who lack the financial capacity to electrify may therefore remain more exposed to carbon taxes, but the magnitude of this burden, and how it depends on revenue recycling, remains quantitatively unknown.

Household electrification entails substantial upfront investment costs, creating a financial barrier to climate adaptation which is unequally distributed across the income distribution. Berg et al. (2024) exploit exogenous variation in unexpected inheritances in Danish administrative data to provide causal evidence that liquidity constraints are a primary obstacle to the adoption of electricity-based durables, especially for poorer households. Consistent with this, Dutch household data reveal an income gradient in heat pump adoption among homeowners, even after controlling for house size and household age, as can be seen in Figure 15 in Appendix A. This income gradient implies that exposure to carbon pricing is unequal along the income distribution. Low-income households who lack the financial capacity to adapt to a green, electricity-driven technology experience a larger increase in energy prices, and therefore energy expenditures. The fact that energy is a necessity good further strengthens this effect as poor households cannot substitute away from energy consumption when relative prices rise.

Against this backdrop, the revenue recycling choice gives rise to a sharp efficiency-equity trade-off. Green subsidies reduce emissions more than lump-sum redistribution for any given tax rate, because they directly lower the cost of clean alternatives. Consequently, a lower peak tax rate suffices to meet a given temperature target, attenuating the distortions to GDP and wages that the tax would otherwise impose. However, because richer households electrify earlier and have higher productivity levels, the benefits of lower electricity prices and reduced wage losses accrue disproportionately to the rich, while the burden falls disproportionately on poor households who cannot adapt. A subsidy may thus be efficiency-enhancing in aggregate while being regressive in its distributional incidence.

In order to quantify to what extent this trade-off materializes, and how the electrification margin shapes it, I construct a quantitative DSGE model with households that are heterogeneous in income, wealth and age in which I conduct counterfactual policy experiments. I model household electrification as an extensive

margin. Households are initially endowed with a brown durable but can choose to invest in a green durable, subject to a large upfront investment costs that slow adoption among the financially constrained. The durable is interpreted as a composite of heating and transport technologies (heat pump + EV for green versus gas boiler + gasoline car for brown). The two durables differ in fuel type used (electricity vs. fossil fuel) and energy efficiency, with the green durable being more efficient.

On the production side, a representative firm produces the final consumption good using a capital-labor composite and energy aggregate, the latter being a CES bundle of electricity and fossil fuels. Fossil fuels are imported, while electricity is produced by profit-maximizing firms combining renewable and fossil-based generation technologies. The carbon tax raises the cost of fossil fuels, inducing substitution towards electricity in final goods production and to renewable technologies in electricity generation itself. The two durable goods are produced by converting the final good into the durable at two technology-specific productivities. Hence, the price of the durable good equals the inverse of its productivity. Lastly, the government implements carbon-tax paths and alternative revenue-recycling schemes subject to a balanced budget. I abstract from climate damages and impose an exogenous temperature objective, which allows for a comparison between alternative revenue recycling instruments.

I calibrate the model to the EU in 2015 in order to quantify the effects of a carbon tax. I find that using tax revenues to subsidize renewable electricity production significantly decreases the carbon tax rate required to limit global warming. With an electricity subsidy, GDP declines are about half the decrease compared to lump-sum transfer recycling. This result does not hold for the electrification subsidy, as direct household energy consumption of poor households, who are policy-induced switchers, has a small effect on total emissions in the calibrated economy.

The attenuated decline of GDP is caused by the significantly lower carbon tax, and hence reduced fossil fuel price increase and an electricity price decrease. The diminished distortions also cause the wage decline to be smaller with an electricity subsidy. Overall the lower electricity price and smaller wage decline benefit households, but do so disproportionately for households who are rich. As these households electrify earlier and more often, they enjoy the lower electricity price more. Furthermore, the proportional wage increase compared to lump-sum is larger for them in absolute terms.

Climate policy imposes the largest welfare losses on low-income, middle-aged households when carbon tax revenues are recycled through green subsidies. This is almost fully driven by the electrification margin, which increases durable and energy expenditures for financially constrained households that are less able to adjust their technology. In contrast, lump-sum redistribution fully offsets the decline in lifetime consumption for low-income households. While this policy requires a higher peak carbon tax rate and therefore leads to larger long-run reductions in GDP, it eliminates the regressivity associated with subsidy recycling.

This paper makes two main contributions. First, I develop a quantitative life-cycle framework that jointly models discrete household electrification subject to adjustment frictions and electricity decarbonization on the supply side, and use it to characterize an efficiency-equity trade-off that cannot be identified from either margin alone. The key insight is that the two margins jointly generate the trade-off: subsidies promoting

electricity decarbonization lower the peak carbon tax required to meet a climate target, reducing aggregate distortions to GDP and wages, but precisely because richer households electrify earlier, the resulting gains from lower electricity prices accrue disproportionately to the wealthy. This interaction generates a trade-off between aggregate efficiency and distributional equity that is invisible to frameworks focusing on either margin in isolation (Ascari et al., 2025; Hochmuth et al., 2025; Kuhn & Schlattmann, 2024; Schlattmann, 2024): the production side determines the magnitude of the distortions and the gains from green subsidies, while the electrification margin determines their incidence. I show quantitatively that this trade-off is sharp — moving tax revenues from lump-sum transfers toward electricity subsidies improves long-run GDP monotonically but worsens welfare outcomes, especially for low-income households, caused by the electrification margin. This result has direct implications for the design of climate policy as the case for green subsidies over transfers breaks down.

Second, I show that introducing household electrification as a discrete, forward-looking investment margin changes the distributional incidence of carbon taxation in a way that is missed by standard models that abstract from either a life-cycle or extensive-margin green technology adoption. Because electrification is an extensive margin, and requires large upfront investment only to be recovered in the future through lower energy expenditures, the relevant household characteristic is not income alone but the interaction of income, wealth, and remaining lifetime horizon. Middle-aged, low-income households bear a disproportionate share of the welfare burden: they are too financially constrained to electrify early to avoid rising fossil fuel costs, yet have enough remaining lifetime to accumulate substantial energy expenditure losses. Young low-income households will eventually electrify; old low-income households are close enough to death that their total exposure is limited. This non-monotonic age profile of welfare losses is obscured in models that abstract from either the life-cycle dimension or the discrete electrification margin, and it implies that standard aggregate inequality statistics understate the concentrated burden on a specific age group. This finding carries direct relevance for the political economy of carbon taxes, as middle-aged households (40-65 years) are often associated with being among the fiercest opponents of carbon taxes (Levi, 2021).

The remainder of the paper is structured as follows. In section 1.1, I relate this paper to the existing literature on the green transition and inequality. In section 2, I discuss the model, that I parametrize in section 3. In section 4, I conduct policy experiments and analyze the results for equity and effectiveness. I conduct some additional analyses in section 5, and conclude in section 6.

1.1 Related literature

A recent surge in interest has led to an increasing number of quantitative papers assessing inequality along the green transition. Benmir & Roman (2022) and Boehl et al. (2024) study how climate policy affects inequality in a setting where energy is modeled as a carbon-emitting input into the production process. In this setting, inequality arises from different channels such as changes in labor income, revenue redistribution or a decrease in future damages. More closely related to the framework developed in this paper is the

literature that explicitly models energy as both a production input and a consumption good. Känzig (2023) examines the effect of carbon pricing in an economy where energy is modeled as a fossil fuel, and serves as an input in the production process and non-homothetically enters the utility function. A carbon tax affects households in two ways, indirectly via reduced economic activity and hence their income, and directly as the households consume energy. In a similar fashion, Fried et al. (2024) use a heterogeneous agent model with non-homothetic preferences for energy to examine the effect of a carbon tax with different rebate instruments on inequality. They find that rebating tax revenues via decreasing distortionary taxes and increasing the progressivity of labor taxes increases welfare more than lump-sum redistribution does.

Other papers make an explicit distinction between green and brown energy sources. In this set-up, climate policy reduces carbon emissions through decarbonization of the production of energy, i.e. electricity. In Ascari et al. (2025), the authors model two distinct energy sources, brown (fossil fuels) and green (renewable). A profit-maximizing energy firm aggregates these two sources of energy in a frictionless manner, based on relative prices and productivity, into a bundle of energy sold to every household and firm. A carbon tax raises the price of fossil fuels, such that the optimal mix contains more green energy. In their model, unlike in my model, all households consume this same optimized bundle, such that the main inequality concern between rich and poor households stems from the non-homotheticity: the share of energy expenditures over income decreases in income. Closely related to their paper, is the paper by Hochmuth et al. (2025) which similarly models a non-homothetic demand for energy from households, and assess the distributional effects of a carbon tax in an economy transitioning towards a carbon-neutral economy. In their paper too, all agents and firms consume the same bundle of energy. They find that both the elasticity of substitution between green and brown energy as well as the degree of non-homotheticity determine the quantitative degree of the inequality effects. In the current paper, I explicitly model both of these types of energy by adding an electrification margin for firms and households, rather than assuming that firms and all households consume the same energy mix. In this set-up, firms and households can consume both fossil fuels directly, and electricity as a composite of renewable and fossil generated electricity similarly modeled as energy in Ascari et al. (2025) and Hochmuth et al. (2025). The goal of climate policy is then to reduce emissions through both increased electrification and decarbonization of electricity production.

A similar distinction between types of energy is made in work examining the spatial margin in this decision by Labrousse & Perdureau (2024). The authors model two sources of energy, brown (fossil fuel) and green (electricity), that are both available for production and consumption. They include a household-specific CES optimized energy bundle based on locational choice (rural versus urban). This implies that a household consumes its own optimal mix of brown and green energy, which is determined by the household's location, and the relative energy prices. Locational choice plays a key role in their model, as households in more rural areas have a stronger preference for fossil fuels than households in urban areas, and therefore are more burdened by a carbon tax. Closely related to this, is the paper by Schlattmann (2024) which incorporates spatial heterogeneity in carbon demand, exploiting the fact that rural households emit more via direct energy demand. In addition to a moving decision, households make a separate decision for a green

or brown car. The paper finds that rural households are more likely to experience a negative net transfer following a carbon tax than urban households. Contrary to the spatial dimension in these two papers, I exploit the fact that electrification requires a large upfront investment, rather than a moving decision.

A paper that also examines this margin is the work by Kuhn & Schlattmann (2024). In their paper, they have a similar set up of modern versus old commitment goods, where the latter are thought of as more polluting. The authors examine the effect of subsidies for switching from the old to the modern commitment good on inequality. In a partial equilibrium setting, they analyze the effect of a subsidy that decreases the purchasing costs for modern commitment goods with different financing instruments on inequality and transition efficiency. Similarly, Crucitti et al. (2024) focuses on heat pump uptake over the life-cycle in a setup where a large upfront investment is required. They analyze the effects of a proportional versus a lump-sum subsidy. They find that as older households usually own bigger houses, they benefit more from proportional subsidies. A similar finding appears in Schlattmann (2024), where richer households own larger houses and therefore benefit more from a lower energy price. In this paper, I abstract from such size complementarities and focus instead on the extensive margin of electrification, the discrete decision to switch technologies, as the primary channel through which climate policy affects inequality. Furthermore, as opposed to the above papers who refrain from modeling an electricity sector and thus only focus on the electrification decision, I include both sides of the transition. Lastly, in the paper by Fried (2026), green technology adoption serves as adaptation against rising temperature variance caused by climate change rather than climate policy. Rich households can adapt to heat pumps that allow for cheaper regulation of the inside temperature in periods of extreme heat or cold. In this paper, I study the policy trade-off that arises from unequal green technology adoption rather than transfers from climate. In the empirical literature, unequal rates of going green along the income and liquid wealth distribution are measured in the Netherlands, Dröes & Van Der Straten (2024) and for OECD countries Ameli & Brandt (2015).

Lastly, this paper relates to the inter-generational comparison of climate policy burden. Fried et al. (2018) examine the distributional effects of different carbon tax revenue recycling instruments between generations by modeling a non-homothetic demand. They find that the generation that carries the biggest welfare burden of a carbon tax is highly conditional on the policy instrument used. They employ a steady state comparison, and evaluate how the future born generation is affected by the use of different revenue redistribution instruments. As opposed to their paper, I examine the short versus long-term effects over the transition, which are caused by the adoption friction rather than a steady state comparison.

2 Model

2.1 Households

Demographics Time is discrete. Households' age is indexed by $j = 0, 1, 2, \dots, J$. After period $J - 1$ the household dies with certainty. A household works for J_w periods, and then retires with certainty. The

economy is populated by a continuum of households with mass one. At birth, households differ with respect to their income $y_{i,0}$, initial wealth $a_{i,0}$ and the durable good they are born with, $\phi_{i,0}$. Households are indexed by i , however for notational convenience this index is dropped in most cases.

Preferences Households maximize expected lifetime utility denoted as

$$U = E \left[\sum_{j=0}^{J-1} \beta^j u(c_j, s_j) + \beta^J w(b) \right], \quad (1)$$

where β is the discount factor, $c > 0$ is consumption of the final consumption good, and s is services derived from energy. Flow utility is a Stone-Geary utility function nested in a CRRA

$$u(c_j, s_j(\phi_j, e_j)) = \frac{1}{1-\sigma} (c_j^\gamma (s_j(\phi_j, e_j) - \bar{s})^{1-\gamma})^{1-\sigma}, \quad (2)$$

where γ denotes the households' relative taste for the final consumption good, $\frac{1}{\sigma}$ is the intertemporal elasticity of substitution, and $\bar{s}$ is a subsistence level for services derived from energy. A subsistence level is included to capture the non-homotheticity in energy demand.¹

The amount of energy services, s , is a function of the level of energy consumption, e , and the type of durable good $\phi \in \{0, 1\}$ (brown vs. green), i.e. $d_\phi = \phi d_G + (1 - \phi) d_B$. The type of the durable good, ϕ , determines the rate at which energy translates into services, such that

$$s(\phi_j, e_j) = \begin{cases} A_{s,G} e_j & : \text{if } \phi' = 1 \implies d = d_G \\ A_{s,B} e_j & : \text{if } \phi' = 0 \implies d = d_B. \end{cases} \quad (3)$$

Akin to the technologies described in Fried (2026), each durable good transforms energy (fossil fuels or electricity in this case) into services of mobility and heating at a different rate. A higher rate implies a more energy efficient durable good. The energy efficiency of the durable good is thus pinned down by its productivity parameters $A_{s,G}$ and $A_{s,B}$ for the green and brown durable good respectively².

A dying household receives utility from bequeathing savings and its durable good. The utility derived from this is denoted as

$$w(b) = \eta \frac{(b + \underline{b})^{1-\sigma}}{1-\sigma} \quad (4)$$

with a strength of the bequest motive η , and where $\underline{b}$ denotes the extent to which bequeathing is a luxury good, similar to De Nardi (2004). Accordingly, the size of the bequest is given by

$$b_t = a'_{J-1,t} + ((1 - \phi'_{J-1,t}) P^{d_B} + \phi'_{J-1,t} P^{d_G}_t). \quad (5)$$

¹Most papers either use a subsistence level to capture this non-homotheticity Wöhrmüller (2024); Schlattmann (2024); Fried et al. (2024) or alternatively the flexible indirect utility function of Boppart (2014), adopted by among others Ascari et al. (2025); Hochmuth et al. (2025). Formally, the former is quasi-homothetic, as it gives linear marginal budget shares in the interior (away from the subsistence level) whereas the latter is fully non-homothetic. Quasi-homotheticity is sufficient to capture that poor households cannot substitute away from energy when the price increases.

²These productivities are calibrated as the coefficient of performance, as defined by the International Energy Agency of the respective durable goods.

Durable good and energy There are two types of durable goods in the economy, a green durable d_G and a brown durable d_B . A durable good converts energy e into services (mobility, heating) at a rate specific to that durable good, $A_{s,G}$ or $A_{s,B}$, and is therefore also interchangeably referred to as a technology³. The two different types of durable goods require different types of energy. A green durable good is fueled with electricity, while a brown durable good requires fossil fuels. Hence, purchasing a green durable good is referred to as electrification.

Each household owns one type of durable good which is denoted using a binary state variable ϕ , such that the durable good owned by the household is given by $d_\phi = \phi d_G + (1 - \phi) d_B$. A newborn household receives either a brown or green durable good from a warm-glow bequest, where the conditional probability of receiving a green durable good is based on its initial income level and determined by the dying cohorts' joint distribution of income and durables. A household can electrify by selling its brown durable good for a scrap value φP^{dB} , and buying a green durable good for $P_t^{dG}(1 - s_{dG})$, where s_{dG} is a green durable subsidy for electrifying households. A household is not allowed to switch from a green to a brown durable good, hence electrification is an absorbing state.

As said, the two goods d_G and d_B differ in the type of energy they require as fuel: the green durable good d_G relies on electricity (green energy), E with price P_t^E , whereas the brown durable good d_B relies on fossil fuels (brown energy), FF , with price $P^{FF} + \tau_{c,t}$. When the government levies a carbon tax $\tau_{c,t}$, this increases the price of fossil fuels. Household aggregate demand for electricity and fossil fuels are given as

$$E_{H,t} = \sum_{j=0}^{J-1} \int \phi'_{j,t} e_{j,t} d\mu_{j,t}, \quad (6)$$

$$FF_{H,t} = \sum_{j=0}^{J-1} \int (1 - \phi'_{j,t}) e_{j,t} d\mu_{j,t}. \quad (7)$$

In case the household chooses to electrify, it can make use of a secured borrowing scheme to pay for the durable, denoted by m , subject to a down-payment constraint, i.e.

$$m'_t \leq \lambda^{LTV} P_t^{dG} (1 - s_{dG,t}) \quad (8)$$

The household is required to repay the loan following an amortization schedule, which implies a fraction

$$\pi(j, t) = \frac{(r_t + r_m)(1 + (r_t + r_m))^{J-j}}{(1 + (r_t + r_m))^{J-j} - 1}, \quad (9)$$

such that next periods loan is equal to $m' = (1 + r_t + r^m - \pi(j, t))m$. The interest rate includes a premium r^m for borrowing by households.

A durable good requires maintenance, which is paid every period and is given by $\delta_\phi = \phi \delta_G + (1 - \phi) \delta_B$.

³This can be thought of as fossil fuels (or electricity) being translated into services of mobility and heating by an electric vehicle (heat pump) and gasoline vehicle (regular boiler)

Income and wealth Households supply their labor inelastically and receive income as an endowment each period. Labor income is denoted as $w_t Y_{j,i}$, where $Y_{i,j}$ denotes labor productivity of a household i at age j , and is given by

$$\log(Y_{j,i}) = \xi(j) + z_{j,i}. \quad (10)$$

such that labor productivity $Y_{j,i}$ consists of a deterministic life-cycle component $\xi(j)$ and an individual component, $z_{j,i}$, representing productivity differences between households. During the first J_w years, the idiosyncratic part of labor productivity $z_{j,i}$ of a household is stochastic and follows an AR(1) process,

$$z_{j,i} = \rho_y z_{j-1,i} + \epsilon_{j,i}. \quad (11)$$

After J_w periods, the household retires. After retirement, it deterministically receives a certain proportion f_{ret} of the income it received in the last period as a pension for the remaining $J - J_w$ periods.

Households can save in a risk-free asset denoted by a . They receive interest rate r_t on their savings every period. Unsecured borrowing is not allowed. Initial wealth is exogenously drawn from a mixture of a degenerate mass-point at zero wealth with probability p_0 and a univariate log-normal wealth distribution with variance $\sigma_{a_0}^2$ that integrates up to the savings bequeathed by dying households in that period.

2.1.1 Recursive formulation household problem

A household is born with initial wealth a_0 , income Y_0 , and durable good ϕ_0 . Thereafter, it solves the following problem during its lifetime, so for $j = 0, 1, 2, \dots, J - 1$, the household solves

$$V_j(a, \phi, m; Y, P_t) = \max\{V_j^{NoAdjust}(a, \phi, m; Y, P_t), V_j^{Electrify}(a, \phi, m; Y, P_t) + \mu_{electrify}\} \quad (12)$$

Where $P_t = \{r_t, w_t, P_t^E, P_t^{FF}, P_t^{dG}, P_t^{dB}, \tau_{c,t}\}$ denotes the price vector, where I drop the time subscript for readability. $\mu_{Adj} < 0$ is a negative preference shock, which represents a type-I extreme value shock. These shocks can be interpreted as a (temporary) dislike to change, distributed iid over time and households. It allows for the adjustment decision to be a probability rather than a discrete choice. The probability of adjustment is a function of the difference between $V_j^{Electrify}(\cdot) - V_j^{NoAdjust}(\cdot)$, but it spreads the mass smoothly which helps for both convergence and calibration purposes. Details can be found in appendix C. The value function for a non-adjuster is given as,

$$V_j^{NoAdjust}(a, \phi, m; Y, P_t) = \max_{a', c, e} u(c, s) + \beta E[V_{j+1}(a', \phi, m'; Y', P_{t+1})], \quad (13)$$

subject to

$$w_t Y + \mathcal{T}_t + (1 + r_t)a \geq c + a' + e[(1 - \phi)(P^{FF} + \tau_{c,t}) + \phi P_t^E] + \delta_\phi P_t^{d\phi} + \pi(j, t)m \quad (14)$$

$$a' \geq 0 \quad (15)$$

$$\pi(j, t) = \frac{(r_t + r_m)(1 + (r_t + r_m))^{J+1-j}}{(1 + (r_t + r_m))^{J+1-j} - 1} \quad (16)$$

$$m' = (1 + (r_t + r_m) - \pi(j, t))m. \quad (17)$$

Equation 14 gives the budget constraint which reads that labor income, transfer income and asset income must exceed or equalize to consumption, savings, energy expenditures, depreciation and any required secured loan repayments. The remaining equations denote an unsecured borrowing constraint, the amortization schedule and the development of the loan repayment.

The value function for a household who chooses to electrify is

$$V_j^{Electrify}(a, \phi, m; Y, P_t) = \max_{a', m', c, e} u(c, s) + \beta E[V_{j+1}(a', \phi' = 1, m'; Y', P_{t+1})], \quad (18)$$

subject to

$$w_t Y + \mathcal{T}_t + (1 + r_t)a + m' \geq c + a' + e\phi' P_t^E + \delta_\phi P_t^{d_\phi} + (P_t^{d_G}(1 - s_{d_G, t}) - \varphi P^{d_B}) \quad (19)$$

$$a' \geq 0, m' \leq \lambda P_t^{d_G}(1 - s_{d_G, t}). \quad (20)$$

Again, equation 19 is the budget constraint, which states that all income plus the taken out loan should exceed at least consumption, savings, energy expenditure, depreciation costs and the purchasing cost of the green durable minus the resale value of the brown.

Lastly $V_J = w(b)$ is as in equation 4.

2.2 Production

There are three sectors on the production side of the economy, one for the final good Y^C , one for electricity Y^E and one for the durable goods Y^{d_G} and Y^{d_B} . Fossil fuels are imported into the economy.

Final consumption good A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity with costs equal to interest rate plus depreciation, wage and the price of electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1 - \psi)^{\frac{1}{\nu}} (A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c})^{\frac{\nu-1}{\nu}} + \psi^{\frac{1}{\nu}} M^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}} \quad (21)$$

where the energy composite M is given as

$$M = \left[\kappa^{\frac{1}{\iota}} (A_{EC} E_C)^{\frac{\iota-1}{\iota}} + (1 - \kappa)^{\frac{1}{\iota}} (A_{FFC} FF_C)^{\frac{\iota-1}{\iota}} \right]^{\frac{\iota}{\iota-1}}. \quad (22)$$

The prices for electricity, fossil fuels, labor and capital are P_t^E , $(P_t^{FF} + \tau_{c,t})$, w_t and $(r_t + \delta)$ respectively.

Electricity There is an electricity producer that inputs capital and fossil fuels and converts it into electricity. Given the capital-intensity of the electricity sector, labor is not used as a production factor.

The profits of the electricity firm are given as follows by an aggregator of renewable and fossil generated electricity multiplied by the price of electricity, minus the costs for inputs, hence

$$\begin{aligned} \Pi(K_{ER}, K_{EF}, FF_E) = & P_E \left(\alpha_R^{\frac{1}{\lambda}} (Y^{ER}(K_{ER}))^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} (Y^{EF}(K_{EF}, FF_E))^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} \\ & - (r_t + \delta)(1 - s_{ER, t})K_{ER} - (r_t + \delta)K_{EF} - (P^{FF} + \tau_{c,t})FF_E \end{aligned} \quad (23)$$

Where $s_{E_R,t}$ denotes the subsidy for the cost of green electricity generation. The production technologies for renewable and fossil generated electricity are

$$Y^{E_R}(K_{E_R}) = A_{K_R,t} K_{E_R} \quad (24)$$

$$Y^{E_F}(K_{E_F}, FF_E) = [\zeta^{\frac{1}{\omega}} (A_{K_F} K_{E_F})^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF_E^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \quad (25)$$

the time subscript for $A_{K_R,t}$ is used to denote that this variable will transition over time in the analyses. In equilibrium, profits are zero.

Durable goods In this economy, the durable goods can be produced by converting the consumption good (with price 1, as it is the numeraire) at some rate A_G, A_B , such that production functions are given as

$$Y^{d_G}(C_G) = A_{G,t} C_G \quad (26)$$

$$Y^{d_B}(C_B) = A_B C_B \quad (27)$$

These imply input demand functions

$$C_G = \frac{1}{A_{G,t}} Y^{d_G} \quad (28)$$

$$C_B = \frac{1}{A_B} Y^{d_B} \quad (29)$$

and prices $P_t^{d_G} = \frac{1}{A_{G,t}}$, $P^{d_B} = \frac{1}{A_B}$. Again, the time subscript refers to a productivity which will transition over time.

2.3 Government

The government runs a balanced budget. It receives carbon tax revenues every period, which it uses for government expenditure, a lump-sum uniform transfer, a subsidy for capital costs to generate renewable energy generation or a subsidy for the costs of electrification, such that every period

$$G_t + \mathcal{T}_t + s_{E_R,t} K_{E_R,t} (r_t + \delta) + s_{d_G,t} P_t^{d_G} \sum_{j=0}^{J-1} \int (\phi' - \phi) d\mu_{j,t} = \tau_{c,t} [FF_{HH,t} + FF_{E,t} + FF_{C,t}] \quad (30)$$

2.4 Equilibrium

Given an exogenous fossil fuel price P^{FF} and a carbon tax $\tau_{c,t}$, equilibrium is defined by a price vector $P_t : \{r_t, w_t, P_t^E, P_t^{FF}, P_t^{d_G}, P^{d_B}\}$ and a set of policies $S_t : \{\tau_{c,t}, G_t, \mathcal{T}_t, s_{E_R,t}, s_{d_G,t}\}$ distribution μ_t over state variables (a, ϕ, m, y, j) such that

1. Given prices, households maximize utility giving rise to value functions $V_j^{NA}(a, \phi, m, y; P_t, S_t)$, $V_j^{electrify}(a, \phi, m, y; P_t, S_t)$ and policy functions $a'_j(a, \phi, m, y; P_t, S_t)$, $c_j(a, \phi, m, y; P_t, S_t)$, $e_j(a, \phi, m, y; P_t, S_t)$, $\phi'_j(a, \phi, m, y; P_t, S_t)$ and $m'_j(a, \phi, m, y; P_t, S_t)$

2. The final-output good firm, electricity firm, and the durable good firms maximize profits
3. All financial wealth bequeathed and all durables bequeathed equals the initial, aggregate wealth of the new-born generation, that is, for wealth

$$\int a'(\mathbf{x})d\mu_{J-1,t} = \int ad\mu_{-1,t}, \quad (31)$$

and durable goods are passed on conditional on initial income level, such that the joint distribution of durables and income is similar in the dying and newborn generation

4. The government budget constraint holds as defined in equation 30.
5. Prices are such that markets clear for

$$\text{Capital:} \quad \sum_{j=0}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \quad (32)$$

$$\text{Labor:} \quad \sum_{j=0}^{J-1} \int y d\mu_{j,t} = L_C \quad (33)$$

$$\text{Gr. Durable:} \quad Y_t^{d_G} = \sum_{j=0}^{J-1} \int \phi \delta_G + (\phi' - \phi) d\mu_{j,t} \quad (34)$$

$$\text{Br. Durable:} \quad Y_t^{d_B} = \sum_{j=0}^{J-1} \int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu_{j,t} \quad (35)$$

$$\text{Electricity:} \quad Y_t^E = \sum_{j=0}^{J-1} \int e\phi' d\mu_{j,t} + E_{C,t} \quad (36)$$

and the aggregate resource constraint, derived in appendix F, clears by Walras law.

3 Parametrization baseline model

In order to assess the quantitative trade-offs behind climate policy during the green transition, I parametrize the economy to match the initial steady state to the pre-transition period. I focus on the European green transition, and calibrate to the year 2015.

Preferences Households are born at age 25, work until 65 and die at 75. This implies J is 50, and J_w is 40. Generations are equal sized. I set the intertemporal elasticity of substitution $1/\sigma$ to $1/2$, which is a standard value in the lifecycle literature. The remaining preferences parameters - $\beta, \gamma, \bar{s}, \eta, \underline{b}$ - are calibrated internally using a simulated method of moments. The key moments targeted in order to discipline these

parameters are the mean savings in the economy, the average energy expenditure over total expenditure, the degree of non-homotheticity in energy expenditures, the change in saving holdings during the retirement phase and inequality in bequests. This is discussed in more detail below.

Durable good and Energy The composite durable converts energy into heat and mobility services. I follow Kuhn & Schlattmann (2024) in defining 2/3 of the composite good as the car and the remaining part as the heating system, according to expenditures on energy for both components.

The rate at which energy is translated into services, $A^{s,G}$ and $A^{s,B}$ for the green and brown durable good respectively are defined as energy efficiency. I calibrate the efficiency levels of the two technologies separately for heat and for mobility and then aggregate them for the composite good.

Energy efficiency is defined as the rate at which a unit of energy, either fossil fuel or electricity, translates into useful energy by the technology at hand. That is, how much useful energy is extracted from using the primary (or secondary for electricity) source of energy. Due to transformation losses or gains at the time of usage, energy services consumed is not the same as final energy demand.

To start with heating, I use energy efficiency estimates from IEA (2022)⁴. They define the concept of useful energy as energy available to satisfy the needs of the user, and refer to this as energy services demand, similar to my model. In their report, they refer to the difference between useful energy and energy used to run the technology as the coefficient of performance (COP), which directly translates into $A_H^{s,G}$ and $A_H^{s,B}$ in my model. They state that heat pumps have a COP of around 4, while there is a huge range from industrial heat pumps having a COP of 3 and district heating networks with large heat pumps showing a COP of 8. I choose to set $A_H^{s,G}$ to 4. Natural gas boilers on the other hand exhibit a COP of around 0.8, hence I set $A_H^{s,B}$ to 0.8.

For mobility, I use estimates from the US department of energy from 2024⁵. They find that for electrical vehicles the energy conversion rate from energy input to useful energy, $A_M^{s,G}$ lies around 0.75⁶. For a gasoline car energy efficiency $A_M^{s,B}$ is lower, as it only translates 0.3 of total energy into useful energy and the remainder is lost as heat. This adds up to 1.82 for $A^{s,G}$, and 0.47 for $A^{s,B}$.

For other parameters, I follow the empirical literature. Following estimates from Schloter (2022), electrical vehicles depreciate faster than gasoline vehicles. Estimates for heat pumps are diverging. I set the depreciation for the green durable good to 2% as opposed to 1% for the brown durable good. Regarding the resale value, I follow a resale value for cars from Gilmore & Lave (2013) of 40% and 0% for heating systems. For the composite good, this implies a resale value of 27%.

The price of fossil fuels is calculated as a composite of the gasoline and the natural gas price, for mobility and heating purposes, converted to kWh. To start with the price of natural gas, from Eurostat⁷, the price

⁴The Future of Heatpumps report

⁵<https://www.energy.gov/eere/vehicles/articles/fofw-1360-sept-16-2024-typical-ev-87-91-efficient-compared-30-conventional>

⁶taking into account cars with and without regenerative breaking

⁷Online data code: nrg_pc_202

Parameter	Description	Value	Internal	Target/source
<i>Households</i>				
<i>Preferences</i>				
β	Discount factor	0.932	yes	See table 2
$1/\sigma$	Intertemporal substitution	1/2	no	Standard value
γ	Taste for consumption	0.913	yes	See table 2
$\bar{s}$	Energy service subsistence level	6.45	yes	See table 2
η	Strength of bequest motive	32.76	yes	See table 2
$\underline{b}$	Luxury of bequest	1.8	yes	See table 2
<i>Durable goods and energy</i>				
$A_{s,G}$	Energy efficiency d_G	1.82	no	See text
$A_{s,B}$	Energy efficiency d_B	0.47	no	See text
δ_G, δ_B	Depreciation	0.02, 0.01	no	Schlöter (2022)
φ	Resale value	0.27	no	Gilmore & Lave (2013)
$1 - \lambda_{LTV}$	Down payment requirement	0.1	no	Own assumption
r_m	Loan premium	0.03	no	Realistic risk premium
μ_{Adj}	Location preference shock	-1.2	yes	See table 2
σ_{Adj}	Scale preference shock	0.32	yes	See table 2
<i>Income and wealth</i>				
ρ_y	Persistence	0.98	no	DNB Household Survey
σ_ϵ	Std. dev. income	0.1	no	DNB Household Survey
σ_{a_0}	Std. dev. wealth at age 0	0.556	no	DNB Household Survey
p_0	Zero-wealth share at age 0	0.39	no	DNB Household Survey
<i>Production</i>				
<i>Electricity</i>				
α_R	Renewable share	0.5	no	Hochmuth et al. (2025)
λ	EoS green and brown electricity	1.8	no	Papageorgiou et al. (2017)
ζ	Capital share E_F	0.3	no	Own assumption
ω	EoS capital and fossil fuels in E_F	0.25	no	Coenen et al. (2024)
A_{KR}	Productivity green capital	14.52	yes	See table 2
A_{KF}	Productivity brown capital	16.26	yes	See table 2
<i>Durable goods</i>				
A_G	Productivity Y^{d_G}	0.86	no	Kuhn & Schlattmann (2024)
A_B	Productivity Y^{d_B}	2.2	no	Price of d_B
<i>Final consumption good</i>				
α_c	Capital share	0.33	no	Literature
ν	EoS capital-labor and energy	0.2	no	Realistically low, see text
ψ	Energy share	0.1	no	Hochmuth et al. (2025)
ι	EoS Electricity and Fossil Fuel	1.5	no	Papageorgiou et al. (2017)
κ	Electricity share	0.2	no	Papageorgiou et al. (2017)
A_{KC}	Productivity capital-labor	1.092	yes	Normalization w
A_{EC}	Productivity electricity	0.018	yes	See table 2
A_{FFC}	Productivity fossil fuels	0.00059	yes	See table 2

Table 1: Parameter values

for one kWh of natural gas is 0.05 euros in Europe, 2015. From the weekly oil bulletin, the price of gasoline (Euro-super95) is 529.7 euros per 1000 liter. Taking as a conversion 9.3 kWh per liter of gasoline, the price of gasoline per kWh in Europe, 2015 is 0.0569 euros. As the composite good consists of 67% mobility purposes, the price of fossil fuels would be 0.0546 euros per kWh. The model unit is defined as 1000 kWh, and normalized by dividing by mean income, such that the price in the model is set to 0.0023.⁸

With respect to the parameters governing the characteristics of secured borrowing for the purchase of a durable good. The minimum down-payment requirement is set to 10% by own assumption in order to set it realistically. Lastly, I set a premium on loans for households to match the fact that these loans often require high interest payments.

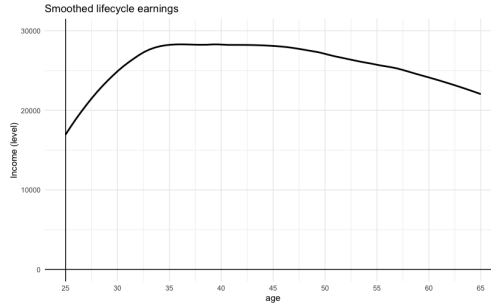


Figure 1: Deterministic life cycle component of productivity, $\xi(j)$

Income and initial wealth Income is determined by the endogenous wage w and exogenous, idiosyncratic productivity $Y_{i,j}$. Productivity is split up in a deterministic lifecycle and a stochastic component. Average productivity is normalized, and together with the wage normalization adds up to an average net labor income of 26033 euros.

The lifecycle component ξ is calibrated using the DNB household survey by regressing the log of net income on age dummies and year fixed effects. The smoothened life cycle pattern of productivity is shown in figure 1. The residual of the regression is then used to calibrate the idiosyncratic, stochastic part $z_{i,j}$. Further details on the data used in calibration of the income process can be found in appendix B.

Initial wealth is a combination of a degenerate distribution at zero wealth and a log-normal distribution over non-zero support, independent of income. I calibrate the fraction of households with a zero level of wealth, p_0 as those households with no or negative financial assets in the data at age 25. With respect to the log-normal distribution, the mean is determined endogenously in order to make sure that the value of assets held by the newborns integrates up to the value bequeathed by the dying households. The variance of

⁸This price pins down the tax in percentage to price per tCO2 unit. For example a tax of 150%. 150% of 0.0023 means 0.00345 per unit of FF, which implies $0.00345 * 26033 = 89.81$ euros. One unit of FF emits 233 kCO2. Hence 4.3 units emit a ton of CO2. So the price is $89 * 4.29 = 381$ euro per tCO2

the log normal distribution, $\sigma_{a_0}^2$ is calculated as the variance of all the observations with a positive amount of financial assets at age 25.

Electricity firm Similar to Hochmuth et al. (2025), I assume α_R to be 0.5. I take estimates for the elasticity of substitution from the literature, namely 1.8 for the EoS between green and brown sources in green electricity generation λ , from Papageorgiou et al. (2017) and 0.25 for the elasticity between capital and fossil fuels in brown electricity production, ω , from Coenen et al. (2024). The remaining parameter, ζ governs the capital share in the production of E_F , and is assumed to be 0.3, in line with literature.

Lastly, The productivity of capital in the electricity sector, for green capital A_{KR} and brown capital A_{KF} are chosen to match (1) the share of renewable electricity and (2) the relative price of electricity over fossil fuels, explained in more detail in the internal calibration section.

The results are robust to changes in ζ ranging from 0.1 to 0.5, as the parameter is relatively inconsequential due to the internal calibration of the productivity to match input shares.

Durable goods production I set the productivity level of the firm producing the brown durable good such that the price of the brown durable good equals nearly half of the mean yearly income, namely 12000 euros. I set the price gap between the green and the brown durable to 1.9, similar to Kuhn & Schlattmann (2024) who calculate this for Germany.

During the transition, I let the price premium of the green durable shrink to zero by increasing the productivity A_G in a linear fashion, such that at time T , $P_T^{dG} = P_T^{dF}$. In order to determine T , I extrapolate the trend of the real price decrease in electrical vehicles and heat pumps over the period 2014-2020.

Final consumption good firm I follow the literature for both the capital-share in the capital-labor composite, and the the elasticity of substitution, ν . With respect to the latter, the literature agrees on a realistically low elasticity of substitution ranging from 0.1 (Hochmuth et al., 2025; Fried et al., 2022) to 0.2 (Olovsson & Vestin, 2023). In order to be on the conservative side in terms of the tax required to decrease industry emissions, but stay within the accepted range, I set ν to 0.2. Part of the results depend this parameter being below 1, which is a well-accepted value in the literature.

Next, with respect to the energy share, ψ , Hassler et al. (2021) show that this parameter is unimportant in this exact problem so long as ν is not close to one. For this parameter, Hochmuth et al. (2025) assume a random value of 0.5, and Ascari et al. (2025) discipline it internally. As the parameter does not move much, and the productivities are calibrated internally conditional on the parameter, I set this value to 10% given low estimates of energy expenditures out of total expenditures in the European economies. To pin down the elasticity of substitution between the different types of fossil, ι , I employ the empirical literature estimate

from Papageorgiou et al. (2017), who estimates both ι and the electricity share κ .

I internally calibrate the rates at which energy is transformed into the final good, so productivities A_{EC} and A_{FFC} in order to match the electricity input share, and the amount of fossil fuels used in industry over total fossil fuels used. Lastly, the productivity of capital-labor is set to normalize the wage to one in the initial steady state.

3.1 Internally calibrated parameters

To discipline 11 parameters that lack a clear data counterpart, I target several implied moments in the simulated data. For identification, I use 9 moments for the initial steady state, and as μ_{adj} and σ_{adj} do not affect the initial steady state, I calibrate 2 moments from the early transition in a second step. All moments are shown in table 2. I match the model implied with their data counterparts using a simulated method of mo-

	Moment	Calculation	Data	Model
γ	Mean energy expenditure share	$\int_{i,j} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}}$	0.125	0.125
$\bar{s}$	Degree of non-homotheticity	$\int_{i,j Q1} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}} / \int_{i,j Q5} \frac{P^{\phi_{i,j}} e_{i,j}}{P^{\phi_{i,j}} e_{i,j} + c_{i,j}}$	1.6	1.59
β	Mean wealth over mean income	$\int_{i,j} a_{i,j} / \int_{i,j} w y_{i,j}$	3.6	4.42
η	Decay of wealth	$\int_i a_{i,J} / \int_i a_{i,J_W}$	0.83	0.78
$\underline{b}$	Wealth inequality at death	$\int_{i D3} a_{i,J} / \int_{i D6} a_{i,J}$	2.84	3.00
A_{KR}	Green electricity share	$\frac{E_R}{E_R + E_F}$	0.175	0.175
A_{KF}	Relative price of electricity	$\frac{P_E}{P_{FF}}$	2.40	2.40
A_{EC}	Input share electricity final good	$\frac{E_C}{E_C + FFC}$	0.34	0.27
A_{FFC}	Final good fossil fuel share	$\frac{FFC}{FFC + FF_{HH} + FF_E}$	0.46	0.47
μ_{Adj}	Share electrified 2017	$\int \phi d\mu_{2017}$	0.012	0.012
σ_{Adj}	Price elasticity d_G 2017	$\frac{\partial \int \phi d\mu_{2017}}{\partial P^d_G} \frac{P^d_G}{\int \phi d\mu_{2017}}$	-1.36	- 1.37

Table 2: Target moments in internal calibration

ments. For the initial steady state, I choose a vector of parameters $\iota = (\gamma, \bar{s}, \beta, \eta, \underline{b}, A_{KR}, A_{KF}, A_{EC}, A_{FFC})$ in order to minimize the objective function

$$\min_{\iota} \sum_{i=1}^9 \omega_i \left(\frac{M_i^{model}(\iota) - M_i^{data}}{M_i^{data}} \right)^2, \quad (37)$$

in which ω_i gives a weight for each moment i .

For the first and second moment, governing the energy expenditure share of households, I use the experimental statistics on income, consumption and wealth from Eurostat for 2015. This dataset uses data from the Household Finance and Consumption Survey. By using household level data, it calculates expenditure shares for a variety of different goods as well as total expenditures. In figure 2 the dotted gray line shows

the energy expenditure share for households in each income quintile in the data. In the internal calibration, I match the average energy expenditure share of 12.56% and the degree of non-homotheticity as measured by the energy expenditure share in the first over the last quintile, equal to 1.6. I calculate the energy expenditure share over total expenditure where the latter is defined as expenditures on all HCIP items. By matching the average, and the degree of non-homotheticity, the energy expenditure per income quintile in the model results in the black line. These two parameters together pin down the behavior targeted in the data as visualized in figure 3. They seem sufficient in getting a good fit over the entire income distribution as shown in figure 2.

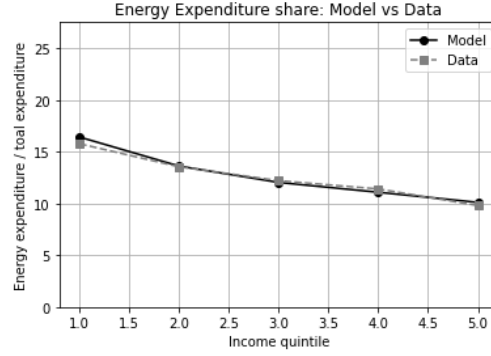


Figure 2: Energy expenditure share per income quintile in Eurostat data (2015) and in the model initial steady state.

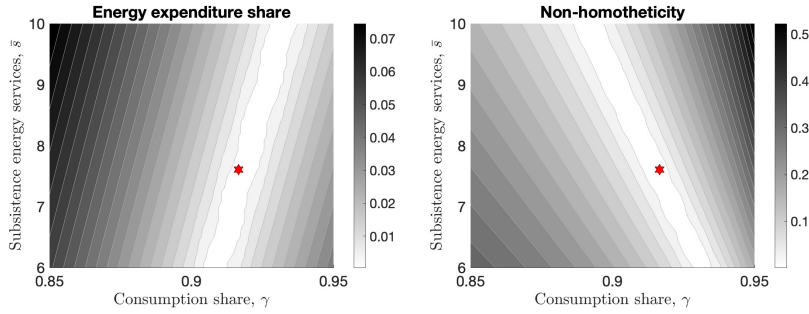


Figure 3: Absolute deviation between simulated model moment and the data target for parameters γ , $\bar{s}$ and A_G and their respective moments the mean energy expenditure share, degree of non-homotheticity in energy expenditures and share of households with a green durable. **Old graphs, replaced soon. Looks the same (just shifts)**

For the moments related to the wealth and income on the household side, i.e. parameters $\beta, \eta, \underline{b}$, I use Dutch micro-data. The first three moments are calculated in the data using the DNB household survey,

waves 2010 - 2020, I discuss this dataset in appendix B. For the discount factor β , I target the ratio of mean household financial net worth (NW) over mean net household income, which in the model is normalized to one. Second, for the strength of the bequest motive, η , I measure the mean decay of net worth in the data as the mean net worth at age 75 over the mean worth at age 65, in order to match the ages of retirement and death in the model. Lastly, to discipline the degree of luxury of bequests, $\underline{b}$ I target the degree of bequest inequality. I compute the mean financial net worth in the third and sixth decile at age 75, and take the ratio as the measure of inequality.

To discipline the productivities in the model, I use aggregate Eurostat data. I set the productivity of the green capital in electricity generation, A_{KR} to target share of electricity that is produced from renewable sources which in Europe, 2015, is 17.5%⁹. To determine the productivity of the brown capital in electricity generation, A_{KF} , I target the relative price of electricity over the price of fossil fuels. As explained above, the price of electricity is 0.0546 per kWh of fossil energy. For the electricity price, I take the DC band price for one kWh in 2015 in Europe, which is 0.131 euros, from Eurostat¹⁰.

The productivities for both energy sources within the production of the final consumption good, A_{EC} and A_{FFC} are disciplined by matching the input share of electricity in its energy mix and the share of fossil fuels used over total fossil fuels used. Using Eurostat aggregate data for the industry excludes the electricity sector but includes all energy used by sectors that conduct other manufacturing and industrial processes. Using natural and manufactured gas, heat, oil and petroleum products, biofossils and solid fossil fuels as a measure of total fossil fuels used in the industry, this implies 34% of all energy used is electricity. To determine the amount of fossil fuels used in the industry, I follow calculations from the energy statistics report from Eurostat¹¹. Taking into account all fossil fuels used in the European economy 46% was used by the industry (defined as manufacturing, storage and services). The remainder consists of fossil fuels used in electricity production (21.8%) and by households for heating and mobility (26.2%). The remaining 6% is used for among other things mining, water supply and construction.

Lastly, in the second calibration step, the two moments are disciplined over the first years of the transition and govern the aggregate rate of and heterogeneity within households that switch from a brown to a green durable. The mean and variance of the extreme value shock, μ_{adj} and σ_{adj} are disciplined by a similar strategy used in Kuhn & Schlattmann (2024), namely by matching the share of households who switched to a green durable and the own-price elasticity of the green durable good. I target these moments after having started the transition. The share of households with an electrical vehicle in 2017 is 1.4% using aggregate statistics from statistics Netherlands (CBS). For the elasticity, I rely on estimates of Fridstrøm & Østli (2021) who use Norwegian data to estimate the price elasticity of cars, among which electrical vehicles. In 2016, they find elasticities of 0.99 and 1.72 for battery electric and plug-in hybrid electric cars. Similarly to the assumption made in Kuhn & Schlattmann (2024), I take the average, which gives an elasticity of -1.36.

⁹Online data code: nrg_ind_ren

¹⁰Online data code: nrg_pc_204

¹¹Energy accounts, 2025

3.2 Baseline model validation

In order to validate the model, I evaluate the baseline switching behavior in the first years of the transition, and I compare inequality in the model and data. These two economy characteristics as they essentially represent the main mechanism (electrification) and the main outcome variable (inequality).

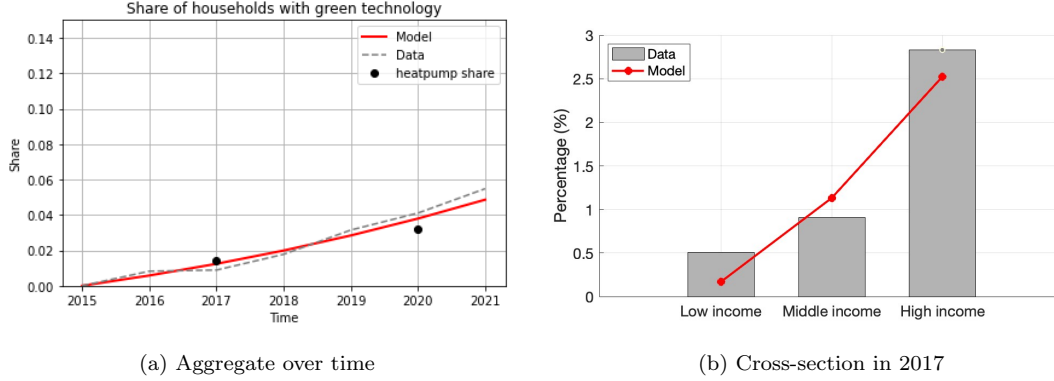


Figure 4: Electrification behavior over the transition

Switching behavior I use publicly available Dutch data from CBS about the share of homeowners with a heatpump, and car owners with an electrical vehicle. As in the model the good is a composite of both, I examine both realized time trends in the aggregate. Figure 4, panel (a) shows a good fit overall of uptake in the first years of the transition. Note that 2015 is the year of introduction so at the start, uptake is zero by default. The aggregate share of the composite good in 2017 is targeted in the internal calibration. The other years are untargeted but still the trend, also in later years.

Additionally I analyze the heterogeneity in electrification between three different income groups. I take a linear combination of the share of households with a heatpump (33%) and the share of households with an electrical vehicle (67%) conditional on income: the lowest and highest quartiles, and the middle half. The percentages in the linear combination follow from the definition in the durable good. For heatpump shares, I use the same micro data as used for the figure in appendix A. For the uptake of electrical vehicles, I use estimates from CBS, who estimate the share of EV-owners in each income quartile.¹² They find that in 2021, from all EV owners, 5.5% is in the lowest income quartile, and 62.4% is in the highest income quartile. I multiply these numbers with the aggregate share in 2017, namely 1.2% to get to the share of EV-owners out of total households in each income group, and divide by the size of the respective income group. The results are shown in figure 4, panel (b). Overall the untargeted distributional uptake of green durables seems a very good qualitative and a sufficient quantitative fit. There is an underestimation of uptake in both the highest and the lowest income group, and a slight overestimation in the middle income group. The electrification

¹²CBS report: who owns electric vehicles

data generated by the model is slightly more linear in income than in the data, where the relation seems a little bit more convex.

Inequality As I compare the distributional welfare effects among different income groups, it is necessary to properly match income inequality. In the EU the income gini ranges from 23.7 to 37.9.¹³ I obtain a gini that is in the upper-end of these bounds, equal to 36.8.

4 Policy Experiments

I solve the model using the algorithms laid out in appendices C - E. The economy starts in the initial steady state, and unexpectedly enters a transition. At the start of the transition, the green durable good becomes available at a price premium. The productivities of production of the green durable good A_G and green electricity A_{KR} increase over the course of the transition.¹⁴ I compare this baseline transition case to several policy experiments where I introduce a carbon tax, also unexpected to households in steady state, at the start of the transition that is phased in linearly.

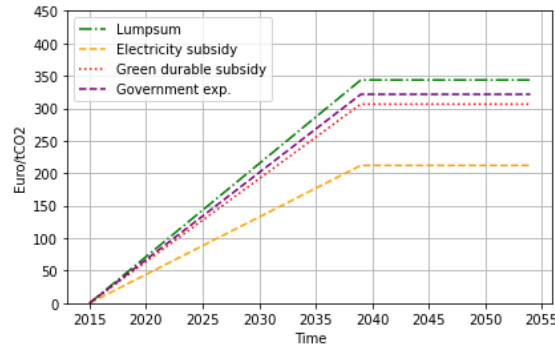


Figure 5: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees Celcius)

In the policy experiments, I increase the carbon tax τ_c from zero in time $t = 0$ to $\bar{\tau}_c^i$ in $T_{\tau_c}^{BL}$ time periods. I choose the exogenous amount of $T_{\tau_c}^{BL} = 25$ time periods. I then compare efficiency (GDP) and distributional welfare outcomes with respect to the baseline (no tax) case by varying the instrument, i , used to recycle the tax revenues. I consider four revenue-recycling instruments: (i) government expenditure, (ii) a uniform lump-sum redistribution, (iii) a subsidy for green electricity production, (iv) a subsidy for

¹³Eurostat, 2015

¹⁴I extrapolate the trends in price developments over the past 10 years and assign the same growth factor for the first 15 years of the transition. For the green durable productivity I make sure it converges to the brown durable's productivity A_B .

electrification of households, I cap the electrification for households at 30%.¹⁵ I compute the tax rate $\bar{\tau}_c^i$ that achieves a maximum temperature increase of 2 degrees Celcius above the industrial level in 2065.¹⁶ I compute the development of temperature as a function of total cumulative emissions, using a model discussed in appendix G.1. The figure for temperature development can be found in appendix G.3. An endogenous tax rate that reaches a given temperature allows for efficiency benefits of green subsidies without explicitly modeling economic damages from temperature increases.¹⁷

Before examining the efficiency and distributional effects, figure 5 shows the auxiliary insight into the endogenous tax paths for each instrument in order to achieve the temperature target. Using subsidies as a recycling instrument, especially a green electricity subsidy, reduces the peak tax rate required to achieve the temperature target. The reason for the lower tax rate with a green electricity subsidy lies in two mechanisms. First, with a subsidy for green electricity, for a given tax rate, the electricity sector substitutes more towards renewable generated electricity. Therefore, electricity production is less emission-intensive. Second, the subsidy for renewable electricity production lowers the electricity price compared to other instruments. Therefore, for a given carbon tax, the final consumption good producer substitutes more towards electricity than fossil fuels within the energy composite, M . The green durable subsidy has a smaller tax-reducing effect due to the fact that households at the margin of switching in the baseline case are mostly poor and have a low overall energy consumption.

4.1 Economic activity over the transition

In order to assess the extent to which the carbon tax distorts economic activity, I assess the development of GDP and prices over the transition. The top panel of figure 6, panel (a) shows the development of GDP over the transition.¹⁸ GDP initially declines after the climate policy introduction for all instruments because higher fossil fuel prices raise firms' production costs. However, the recovery of GDP after the initial dip is highly dependent on the instrument used. The electricity subsidy mitigates GDP losses from the carbon tax most as it dampens distortions arising from higher energy prices.

In equilibrium, climate policy increases the interest rate (figure 6, panel (b)) and decreases the wage (figure 6, panel (c)). The increase in the interest rate is caused by a shift from fossil energy imports (and thus final consumption good exports) to electricity production, which is more capital-intensive than the

¹⁵I do this as the revenue including the firms' tax payments exceeds the amount that can be given out as subsidies (that is tax revenues exceed all switching costs in early periods even if everyone where to switch). The 30% is in line with most taxes given out in Europe. In the default case, the remainder is spent on government expenditure.

¹⁶As the economy is not in net-zero at this point, the underlying assumption is that some carbon capture technology keeps the temperature at 2 degrees Celcius.

¹⁷Alternatively, I could set an exogenous tax rate and compute climate damages. In this case, climate damages would be lower with a subsidy rather than lump-sum redistribution or government expenditure due to lower emissions, which reduces GDP declines. In the current case I account for this by allowing for a lower tax rate under subsidy redistribution, also leading to lower GDP declines.

¹⁸Discontinuities in the graph are explained by the phasing in of increased renewable electricity production for 15 periods (2030), and the tax for 25 periods (2040).

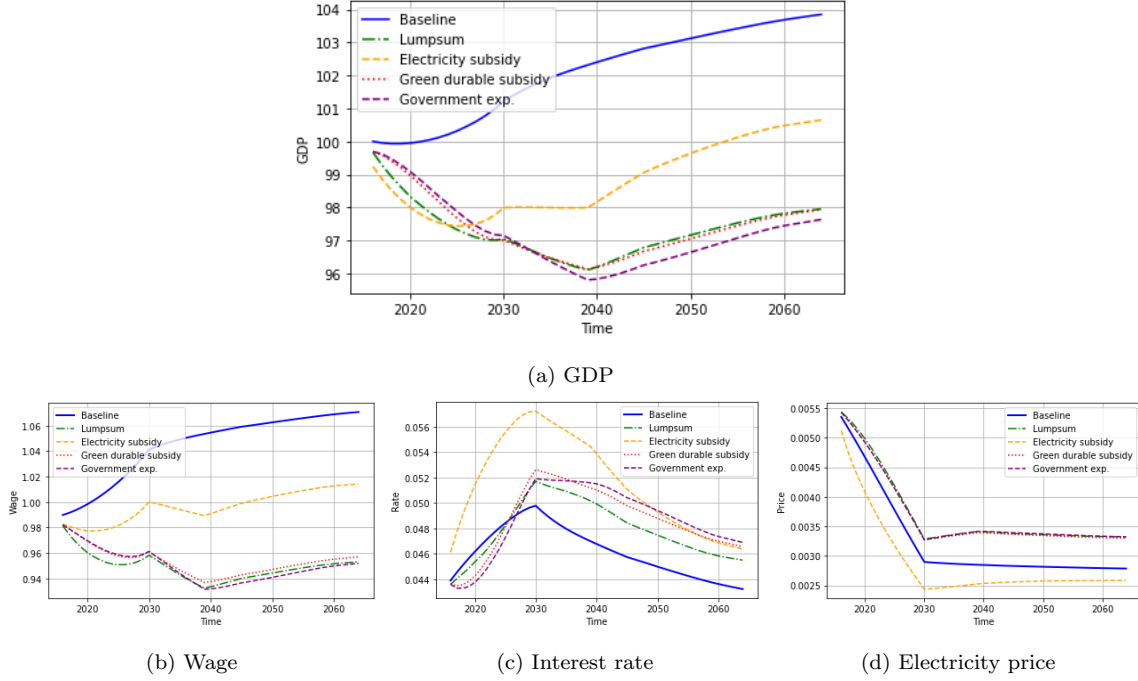


Figure 6: GDP (top) and prices (bottom) over the course of the transition

production of the final consumption good. This increase in capital demand pushes the interest rate up. This effect is amplified in the short run, as households spend their financial wealth on green technology adoption, and would otherwise have lower savings.

Contrary to the interest rate, the wage decreases. As labor is only used in producing the final consumption good, the wage is fully determined within that sector. The price of the optimized energy composite, M , used in the production of the final consumption good increases for all recycling instruments. Because the capital-labor composite is complementary to the energy composite (elasticity of substitution, $\nu < 1$), the wage drops. The muted wage drop in the case of the electricity subsidy is attributable to the muted increase in the price of the energy composite M . The price of M increases less with an electricity subsidy for two reasons, first the cost of fossil fuels is lower, following the lower endogenous tax rate estimated above, and second the lower electricity price as visible in figure 6, panel (d). The same reasoning holds for the muted decrease in GDP in the electricity subsidy case.

4.2 Distributional Welfare Effects and the Electrification Margin

In figure 7 I show the consumption equivalence for the different recycling instruments for the average household in a given age and productivity group (j, z) for all households alive when the policy is introduced. With

the exception of uniform lump-sum transfers as the recycling instrument, all agents are worse off from the carbon tax. In the case of lump-sum transfers, low-income households gain from the policy due to the fact that the increase in their net disposable income, i.e. the transfers and labor and asset income decreases, exceeds increased expenditures for them.

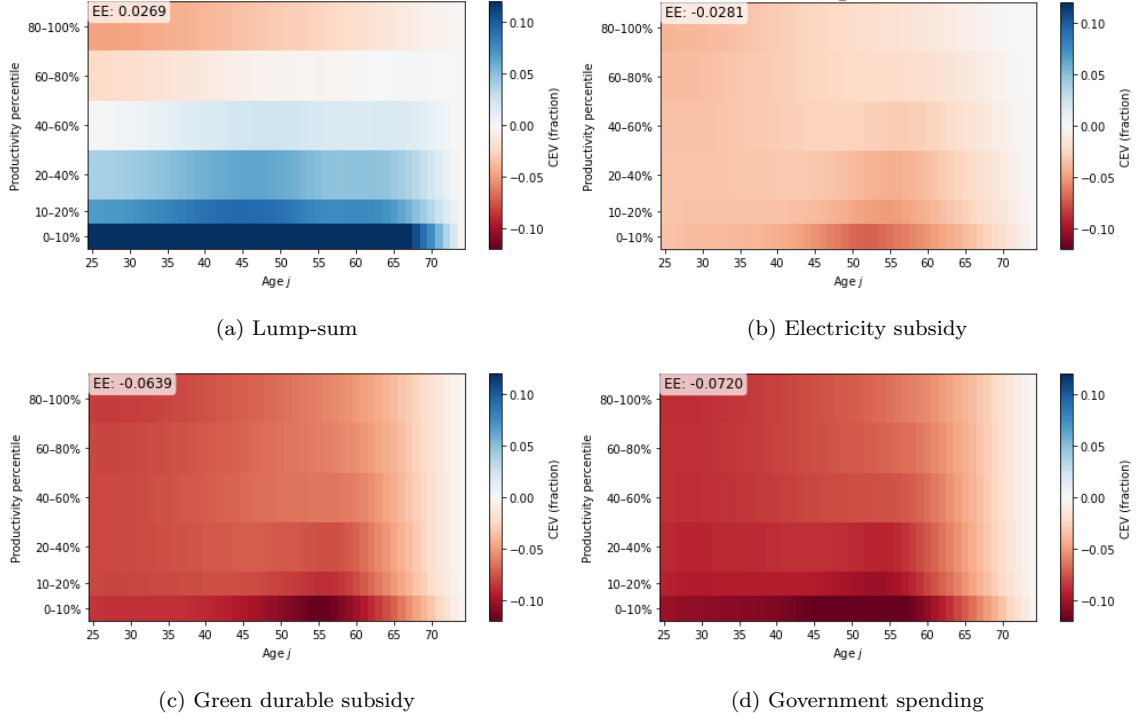


Figure 7: Consumption equivalence across the age–income space for different recycling instruments. Values indicate the percentage change in expenditure in the baseline required for households to be indifferent between the baseline and policy scenario.

In all three other cases all agents are worse off with carbon tax, and poor middle-aged households suffer the largest welfare loss. The electrification margin is the key driver of the high welfare loss for this group. The loss stems from increased expenditures, either through increased spending on the durable good when the household electrifies in order to adapt to the carbon tax and avoid high energy expenditures, or through higher energy costs when electrification does not take place. I elaborate on these expenditures below, as well as on why a lump-sum transfer back to households erodes the welfare loss caused by increased expenditures for this group in the next section. Before I do so, I first compare the two types of green subsidies among each other.

Subsidies increase welfare compared to the use of tax revenues for government expenditure. When the subsidy is directed at renewable electricity, the welfare losses are smaller overall compared to an Green durable

subsidy. The significant welfare difference between subsidy types is attributable to the attenuated negative general equilibrium effects in the electricity subsidy case. As explained above, the carbon tax and price of electricity are significantly lower following the electricity subsidy, therefore wages decrease significantly less for the same carbon tax. Overall, these two effects lead to electricity subsidies being superior over an green durable subsidy in terms of welfare effects.¹⁹ However, the lower decrease in wages and the drop in electricity prices mostly benefit the rich, high productivity households who electrify early. Put differently, the welfare gains of a green electricity subsidy are mostly concentrated among the rich households.

4.2.1 Mechanisms

To understand the mechanisms underlying the distributional results, I study two related questions. First, why does lump-sum redistribution benefit low-income households? Second, to what extent and how does the electrification margin shape the distributional incidence of climate policy? I address these questions by analyzing the change in expected lifetime consumption, denoted by ΔLC , following the introduction of the policy. The corresponding welfare effect of the consumption change depends on sign and size of the change, the timing of the change as welfare effects are discounted, the degree to which the change is spread out over time, and the initial level of consumption. The latter two are caused by the concavity of the value function. For these reasons, the overall magnitude of the change in consumption is nonlinearly translated in the magnitude of welfare effects. However, the sign of the consumption change exhibits a one-to-one relation with the sign of the consumption equivalent. Furthermore, within income groups, the size remains indicative. Lastly, the largest advantage of analyzing ΔLC is that I can use the budget constraint to decompose changes in ΔLC into five different components, that I formally define in proposition 1.

Proposition 1. *Consider household i , who is age $j_{0,i}$ at the time of the policy introduction. Define the remaining lifetime horizon as $S_i \equiv J - 1 - j_{0,i}$. For $s = 0, 1, \dots, S_i$, define household i 's lifetime consumption object as*

$$LC_i \equiv E \sum_{s=0}^{S_i} c_{i,j_{0,i}+s,s} + a_{i,J},$$

and define the five components

$$\begin{aligned} \text{Transfers}_i &\equiv \sum_{s=0}^{S_i} \mathcal{T}_s, & \text{LaborInc}_i &\equiv E \sum_{s=0}^{S_i} w_s y_{i,j_{0,i}+s}, & \text{AssetInc}_i &\equiv E \sum_{s=1}^{S_i} r_s a_{i,j_{0,i}+s,s}, \\ \text{EnergyExp}_i &\equiv E \sum_{s=0}^{S_i} EE_{i,j_{0,i}+s,s}, & \text{DurableExp}_i &\equiv E \sum_{s=0}^{S_i} DE_{i,j_{0,i}+s,s}, \end{aligned}$$

¹⁹The combination of a green durable subsidy and electricity subsidy does not improve welfare much beyond what the electricity subsidy compared to the green durable subsidy does. I discuss this case as an additional analyses below.

where

$$EE_{i,j,t} \equiv P_{\phi_{i,j,t},t} e_{i,j,t},$$

$$DE_{i,j,t} \equiv \delta_{\phi_{i,j,t},t} + f(j) m_{i,j,t} - (\phi'_{i,j,t} - \phi_{i,j,t}) (m'_{i,j,t} - P_t^G + \phi P^B).$$

Then the change in household i 's expected lifetime consumption relative to the baseline admits the decomposition

$$\Delta LC_i = \Delta \text{Transfers}_i + \Delta \text{LaborInc}_i + \Delta \text{AssetInc}_i - \Delta \text{EnergyExp}_i - \Delta \text{DurableExp}_i. \quad (38)$$

Proof. See appendix H. □

Why is lump-sum beneficial for low-income households? In order to understand why lump-sum is so beneficial for poor households despite the higher tax rate on fossil fuels compared to the electricity subsidy, I plot the expected lifetime consumption change, ΔLC_i , as well as the five different components as a percentage share of total expected lifetime consumption absent any policy, for three different income groups (conditional on age) in figure 8. The denominator is thus the same in both panels.

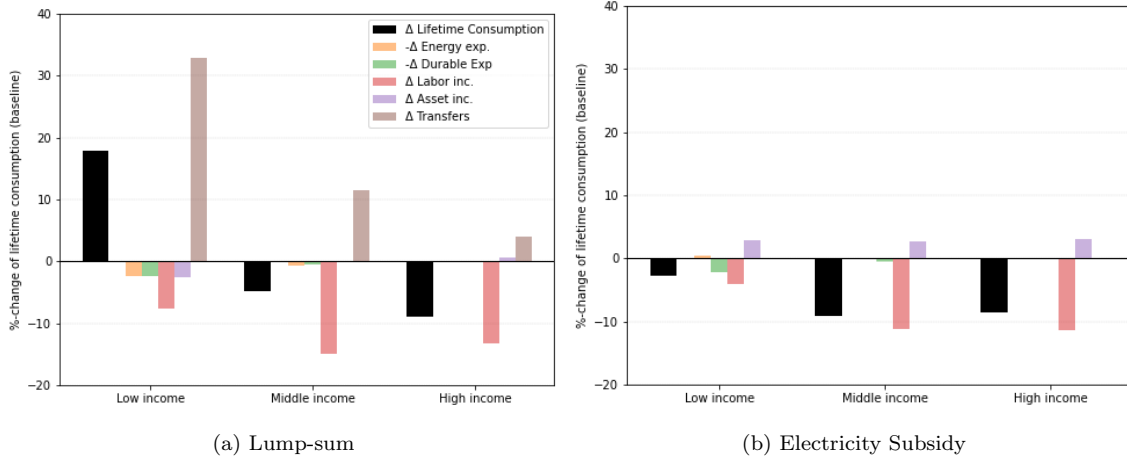


Figure 8: Decomposition of the change in expected lifetime consumption following policy introduction. Low-income refers to the lowest decile (conditional on age), high-income refers to the highest decile.

The largest part of the decrease in lifetime consumption comes from the decrease in wages. Comparing within income group variation between both instruments shows that, as expected, electricity subsidies attenuate the wage decline and thus the loss in labor income. However, when focusing on between income group differences within instrument, a wage decline affects labor income of high earners by a larger magnitude. This difference is driven by the fact that a proportional wage decrease affects high productivity

individuals more.²⁰ Therefore, for low-income households, the lump-sum transfer is larger in magnitude than the combined effect of all other (negative) components, whereas for other income groups the overall change in lifetime consumption remains negative. This sign difference between income groups disappears for the electricity subsidy case, where low-income households have a small negative change in expected lifetime consumption. While low-income households are worse off with electricity subsidy compared to lump-sum distribution, high-income households are slightly better off with electricity subsidies.

Judging from figure 8, low-income households seem to lose little consumption in the electricity subsidy case. However, the results in figure 8 are averaging out any life-cycle effects and hide the within-income group heterogeneity. In order to understand why especially the middle-aged households in the lowest income groups are most burdened, I disaggregate the consumption response for the lowest income groups further along the age dimension when quantifying the effect of the electrification margin.

To what extent and how does the electrification margin affect the distributional incidence? To quantify the role of the electrification margin in shaping the distributional welfare effects of the carbon tax, I compare the baseline consumption equivalent to one from a counterfactual economy in which all households are endowed with a green durable. In this counterfactual economy, the extensive electrification margin is effectively removed, allowing me to isolate the welfare consequences after a policy introduction caused by the electrification margin. I define the state-dependent welfare effect of the electrification margin as

$$\Delta CE^{Elec}(j, z) = CE_{t=1}(j, z) - CE_{t=1}^{noElec}(j, z),$$

where thus in the *noElec* case all households are endowed with green durables. I plot this state-dependent welfare effect in Figure 9. A positive value implies that the electrification margin improves the welfare impact of the policy, while a negative value indicates that it causes a welfare loss following the policy introduction.

In all cases except lump-sum redistribution, the electrification margin causes households to be significantly worse off from a carbon tax. On the contrary, when using lump-sum redistribution the existence of the electrification friction actually makes poor households a little bit better off. This happens because with electrification margin, at the start of the transition, tax revenues and therefore the lump-sum transfer is higher. However in all other cases, the margin makes that poor, middle-aged households suffer disproportionately from climate policy. In order to understand why the electrification margins affects these groups most, I analyze the age heterogeneity in durable and energy expenditure, as defined in proposition 1, as this heterogeneity explains the red dark triangle in figure 9. In figure 10, I plot the changes in durable expenditures, energy expenditures and their sum for the lowest income group, paired age cohorts, for lump-sum and electricity subsidy recycling.

²⁰This is inherit to a standard income process, where wage decreases are uniformly distributed. Others have assumed differently, however, see for example Känzig (2023) or Labrousse & Perdureau (2024). If it were assumed that low productivity households are more affected by wage drops or labor market loosening, this would negatively affect the case for lump-sum compared to electricity subsidies.

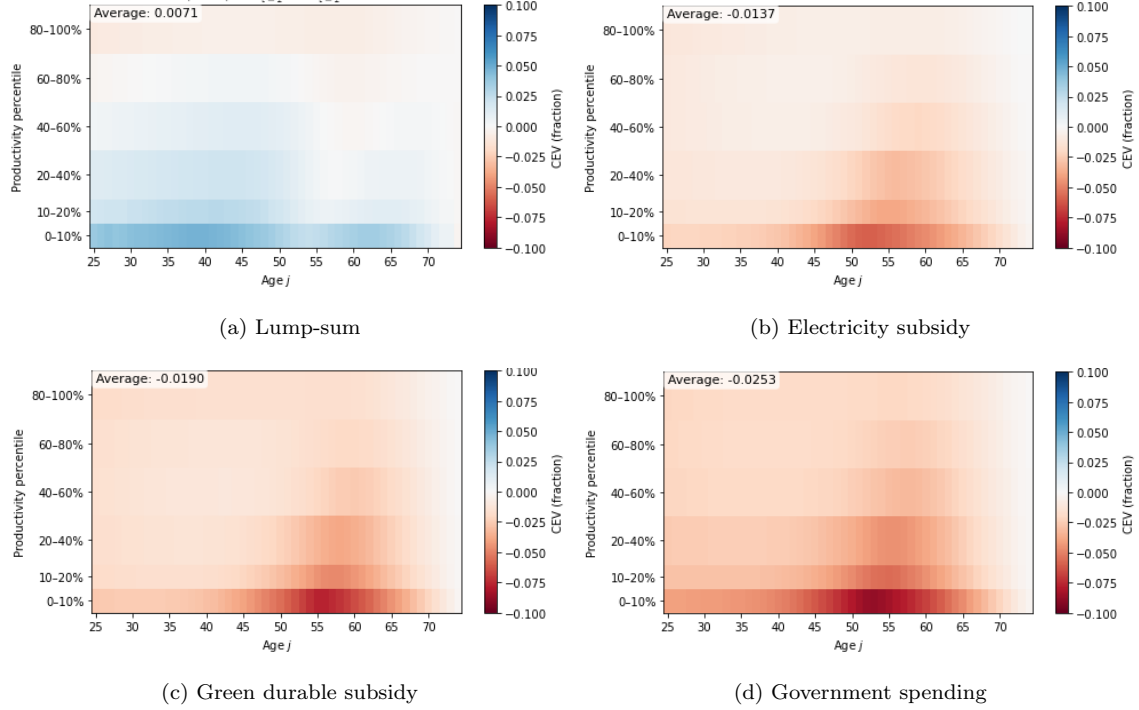


Figure 9: State-space heatmap of $\Delta CE^{Elec}(j, z)$ for different recycling instruments

In figure 10, it is useful to separate three groups of households: (1) the young (roughly ages 25–35) households who would have electrified in any case, (2) the old (roughly aged > 53 years old) households who are not electrifying in any case, and (3) the switchers; the middle-aged (roughly ages 35–53) households who would not have electrified without the carbon tax in place, but will do so with carbon tax. In group 1, the young households incur a small increase in durable expenditures as they electrify earlier with the policy in place, when P_t^G is still higher. Furthermore, in case of lump-sum redistribution, as the electricity price rises somewhat, so do their energy expenditures. As a result, the effect on lifetime consumption via increased expenditures is limited for this group. In group 2, since these households are too close to the end of life to recover the fixed electrification costs, they optimally stick with the brown durable, and consequently face a persistent increase in energy expenditures as fossil fuel prices rise. In group 3, the group of policy-induced switchers, electrification requires paying substantial upfront costs, generating a pronounced increase in durable expenditures. Whether these households are compensated through lower energy expenditures depends on the recycling instrument. Under lump-sum redistribution, electricity prices rise, so switching households experience both higher durable expenditures and slightly higher energy expenditures. Under an electricity subsidy, electricity prices fall, which partly offsets the switching costs through lower future energy expenditures. Overall, changes in energy and durable expenditures affect lifetime consumption of the

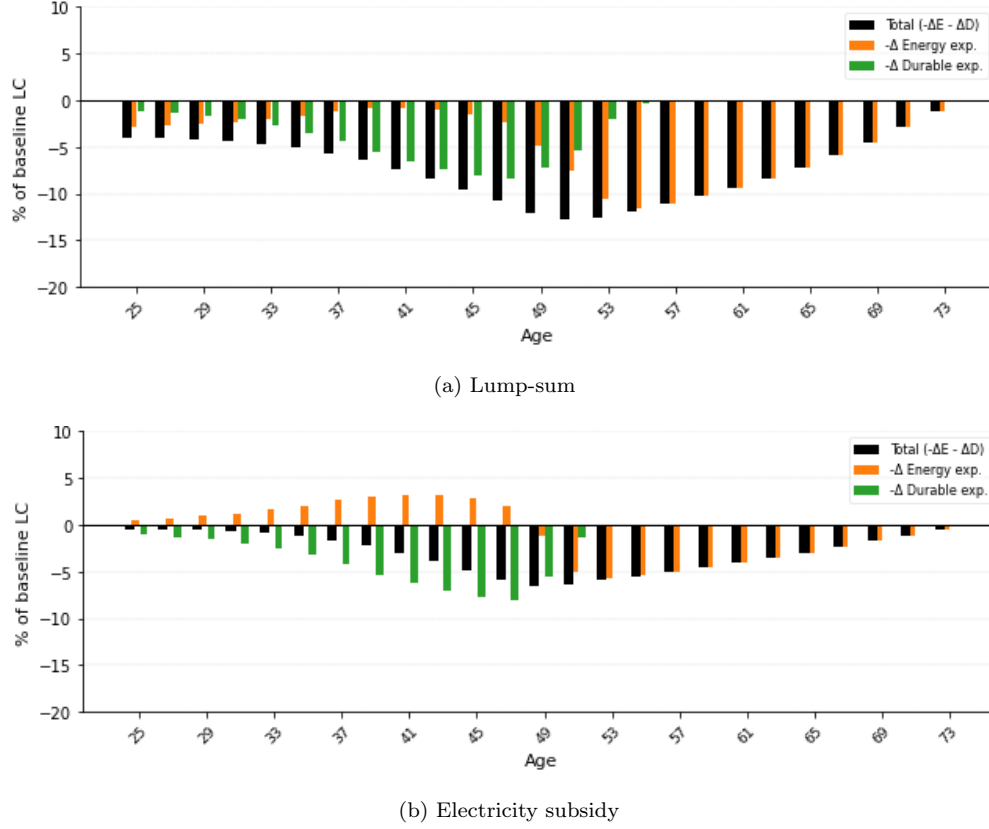


Figure 10: Heterogeneous expenditure responses across the age space for $z = 0$.

middle-aged (with the peak around ages 49-57, depending on the instrument) households most, which are the ones on the margin of electrifying and not electrifying with the carbon tax in place.²¹

A carbon tax thus disproportionately raises durable and energy expenditures for middle-aged to old, low-income households, whether they switch technologies or remain on fossil fuels. Under lump-sum recycling, these expenditure burdens are larger since both fossil fuel and electricity prices rise. Under the electricity subsidy, switchers benefit from lower electricity prices and non-switchers face a smaller fossil fuel price increase, so expenditure burdens are lower. Nevertheless, the financial relief for poor households is larger with lump-sum redistribution due to the size of the transfer compared to their lifetime consumption. In short, the subsidy better targets expenditure increases from the electrification margin as well as income

²¹Similar figures for the green durable subsidy and government expenditure can be found in Appendix I.1. The figure for government spending explains the triangle with the same mechanisms but increased magnitudes. The figure for green durable subsidy diminishes durable spending, which explains why the triangle in figure 9 panel (c) shifts to the right and thus the margin is especially salient for never-switchers.

decreases, but this mainly benefits rich households while lump-sum transfers provide greater net income support to poor households.

The case without electrification In contrast to the findings of Ascari et al. (2025), who show that a carbon tax combined with an electricity subsidy is regressive, I would not find regressivity with an electricity subsidy but without the electrification margin. In their framework, regressivity arises from non-homothetic energy demand and the fact that the price of energy increases even when electricity is subsidized. The key difference is that, unlike in their setting, the price of electricity in my economy decreases when the policy is implemented with an electricity subsidy. As a result, even with non-homothetic energy demand, climate policy with revenue recycling through electricity subsidies is progressive. The reason for the declining electricity price is the assumption of continuous fuel switching of the final consumption good firm, that thus also uses fossil fuel. Because this firm pays the carbon tax on its fossil fuel usage, tax revenues increase substantially. These additional revenues finance the renewable electricity subsidy, and the resulting subsidy-induced reduction in electricity prices more than offsets the upward pressure from the carbon tax on fossil fuels. Consequently, electricity prices fall, leading to progressive policy effects.

4.2.2 Combining lump-sum and electricity subsidies

In this section, I analyze whether combining electricity subsidies with lump-sum transfers can mitigate the trade-off between efficiency and equity. To assess this, I consider linear combinations of the two instruments, varying the share of tax revenues allocated to each. Table 3 reveals the central trade-off in this paper: subsidizing green electricity will relieve GDP distortions caused by higher energy prices, but as the resulting benefits mainly accrue to richer households, aggregate welfare losses and inequality increase. Both the efficiency and aggregate welfare results vary approximately linearly in the share of revenues allocated to lump-sum transfers versus electricity subsidies. Increasing the weight on lump-sum redistribution monotonically raises GDP losses while improving welfare (especially for low-income households), whereas shifting revenues toward electricity subsidies reduces GDP losses but worsens distributional outcomes.

% Tax Rev. Subsidy (LS)	Future GDP Decrease (%)	Current EE Aggregate (%)	Current EE low-inc. (%)
100 (0)	-3.37	-2.81	-4.41
75 (25)	-3.74	-1.81	-0.11
50 (50)	-4.21	-0.60	3.11
25 (75)	-4.85	0.84	8.10
0 (100)	-5.80	2.69	14.83

Table 3: Policy trade-off between subsidy financing and lump-sum transfers

The linearity implies that the two instruments operate through largely separable channels. Electricity

subsidies primarily affect the economy through general equilibrium effects—lower electricity prices which reduce production distortions—while lump-sum transfers operate through direct redistribution. Because these channels interact only weakly, combining the instruments does not generate additional complementarities beyond a convex combination of their individual effects. As a result, policy mixes trace out a clear efficiency–equity trade-off. Moving toward electricity subsidies improves aggregate efficiency but disproportionately benefits higher-income households who electrify early and therefore gain from lower electricity prices. In contrast, increasing the lump-sum component redistributes resources toward poor households, more than offsetting their higher energy expenditures, but at the cost of larger distortions from a higher carbon tax. These results imply there is no obvious interior optimum in this setting. The division of revenues remains a normative question that depends on the calibration of the welfare function as well as the discount factor used for future generations.

4.3 Robustness and validity

A concern with respect to the validity of the results comes from the fact that economic damages from temperature increases are not modeled, and this might trouble the comparison between the baseline and the policy scenarios. The comparison between the different policy instruments is not affected by this, as the carbon tax path is estimated such that temperature changes are not diverging. Therefore, conclusions drawn are only applicable to comparisons across the policy instruments themselves, rather than to comparisons between any policy scenario and the baseline.

Moreover, four assumptions are important for the results and are natural candidates for robustness checks. First, idiosyncratic risk burdens households but is partly offset by lump-sum redistribution, as deterministic transfers facilitate consumption smoothing. Removing idiosyncratic risk while keeping the income grid fixed, so that income differences reflect only ex-ante heterogeneity, does not significantly affect the distributional welfare results; Second, with respect to the elasticity of substitution between energy and the capital-labor composite in the final consumption good sector, ν , as this determines the wage decrease arising from the tax. I find that the level of distortions are robust against increasing the elasticity of substitution, when keeping tax rates constant; Third, the elasticity of substitution between renewable and fossil generated electricity, λ , as this largely governs the tax rate required to limit temperature increases for different instruments. I find that the height of the tax rate is sensitive to changes of the elasticity of substitution between brown and green electricity within the sector, however there are no interaction effects with the instruments used; and last the minimal down-payment required for the purchase of the green durable good, λ^{LTV} as this governs the severity of financing frictions. The results are robust when loosening the borrowing constraint. Households who do not electrify early are not affected by loosening the borrowing constraint as they would still need to pay an interest rate on their borrowing, it is not profitable to adopt earlier. The figures supporting these results as well as a more extensive description of the results can be found in appendix J.

5 Additional analyses

In this section I analyze extensions to the climate policy design decisions. I start by adjusting the time-frame in which the carbon tax is phased in and assess the results of front-loading the carbon tax on the aforementioned equity-efficiency trade-off. In a second extension I examine the effect of combining the two green subsidies.

5.1 Front-loading the carbon tax

Front-loading the carbon tax offers the advantage that the growth of the cumulative carbon stock reduces earlier, and therefore a lower tax rate can achieve the same climate target. Similarly to how $\bar{\tau}_c^i$ in $T_{\tau_c}^{BL} = 25$ was estimated, I find the endogenous tax rate when the tax is implemented in $T_{\tau_c}^{FL} = 15$ time periods for each instrument. In figure 11 I plot the results of the endogenous tax rates when front-loading.

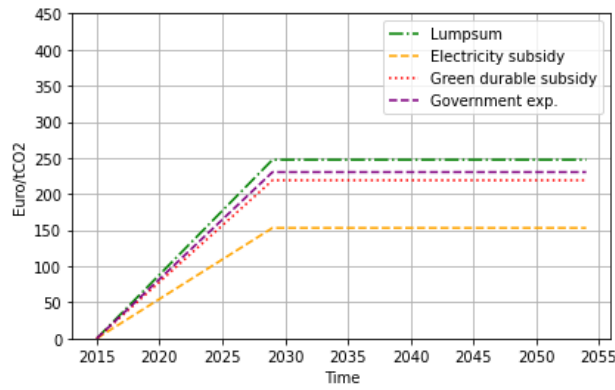


Figure 11: Carbon tax path by revenue-recycling instrument to meet the exogenous temperature target (2 degrees celcius) with front-loading

The qualitative order of peak tax rates for the instruments remains the same, however compared to the baseline case, all peak tax rates are significantly lower in the long run. Therefore, in the new steady state after the transition has completed, GDP and wages will be higher than in the baseline case. However, due to front-loading, in the short-run the carbon tax is higher than in the baseline case. And since in the short-run (up to $t = 15$) the productivity gains in the electricity sector have not yet fully materialized, there is a large short-run GDP and wage dip in the front-loading case. The latter leads to a larger negative aggregate welfare burden for generations alive when the policy is introduced. I visualize this timing trade-off for each instrument in figure 12.

The left panel shows for each instrument, the GDP decline in the final steady state (x-axis) and the aggregate current welfare loss for alive generations (y-axis) for the baseline (BL) and front-loaded (FL) tax path. The right-hand panel shows the equivalence but with the welfare loss of the lowest income group rather

than the aggregate. A downward sloping curve between FL and BL implies a timing trade-off: increasing the speed of phasing in raises GDP, but increases (decreases) the current welfare loss (gain). The solid lines show the trade-offs for the case with electrification margin, the dashed lines show the results when the electrification margin is excluded.

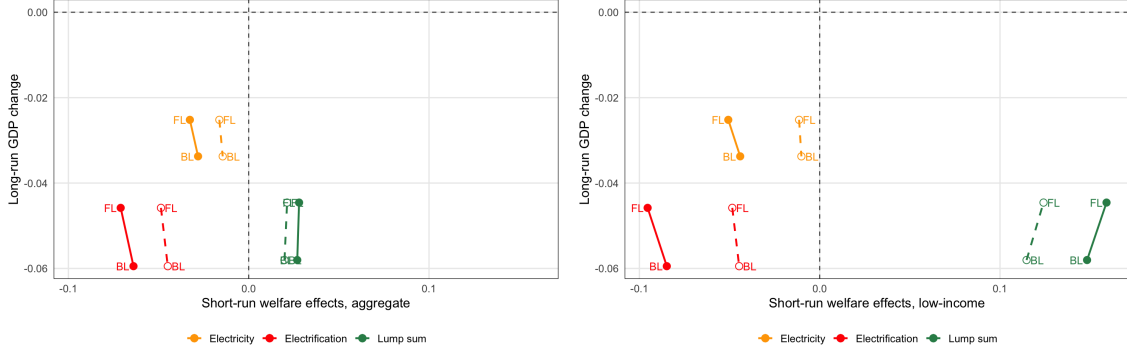


Figure 12: GDP decline in the long run versus aggregate welfare change now (left) and low-income group welfare change now (right) for the baseline (BL) and front-loaded tax (FL) case. Solid line shows results with electrification margin, dashed line shows results without electrification margin.

The figure highlights various results. First of all, for all instruments, the long run GDP decrease compared to no policy is smaller when front-loading the tax, which is caused by the lower tax rate (figure 11 versus figure 5). Second, what was already established above but is well visible in this figure, is that the electrification margin affects welfare negatively after a carbon tax introduction (with non-lump-sum recycling), and this effect is significantly larger in magnitude for low-income households. Looking at the difference between front- and back-loading the tax for welfare outcomes, front-loading increases the magnitude of the welfare effect: for subsidy recycling welfare becomes more negative, while with lump-sum it becomes more positive. These welfare effects are larger due to the electrification margin, especially for the poorest households. This implies that front-loading hurts the low-income households the most, and this is caused by the electrification margin. Basically, as the low-income households cannot adopt green technologies (early), the higher short-run tax rate hurts these households more. Lump-sum redistribution alleviates the timing trade-off.

5.2 Combining green subsidies

In the above analyses the residual of the total budget left after capping the electrification subsidy to 30% was used for government spending. Government spending increases GDP but does not affect households and therefore welfare. In this section I analyze the results for GDP, prices and welfare when I combine both subsidies. I do this by subsidizing up to 30% of the price of the green technology when a household electrifies, and recycle the residual tax revenue using an electricity subsidy. The resulting carbon tax amounts to 207

euro/tCO₂, almost exactly equal to the tax in the case of pure electricity subsidy recycling.

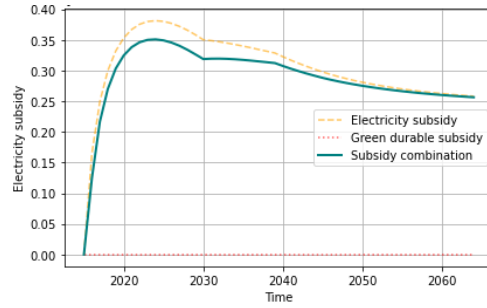


Figure 13: Electricity subsidy with full electricity subsidy, electrification subsidy (and government spending) and a combination of the two subsidies

Figure 13 shows that the electricity subsidy is not reduced by much when the first 30% of all green technology purchases are subsidized compared to the case where there is no electrification subsidy. The differential completely disappears towards the end of the transition, when most households have electrified. The implication of this is that in terms of GDP and prices, the combination of the two subsidies is very similar to the case of full electricity subsidies. Figure 21 in Appendix I.3 shows this.

The economic activity results thus hardly differ from the results of a pure electricity-based subsidy. This finding extends to the distributional incidence of the welfare results, shown in figure 14. Overall, welfare improves by approximately 0.4 percentage points. However, the policy remains regressive. Overall, this option offers an improvement over both individual subsidies. It offers the advantage of low wage declines and lower costs of adaptation simultaneously. It does not resolve the significant negative burden for households who are unable to electrify: the poor, middle-aged households who see their energy expenditures rise.

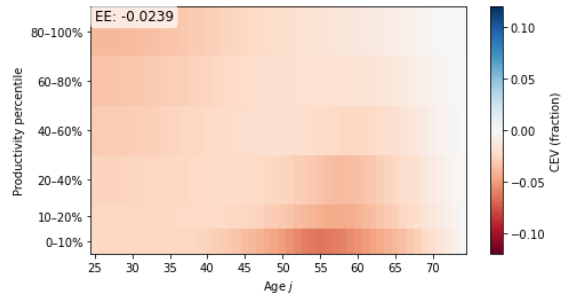


Figure 14: State-space heatmap of the expenditure equivalence across the age-income space for the combination of subsidies

6 Conclusion

In this paper I studied the effects of a carbon tax on efficiency and inequality in an economy containing both sides of the energy transition: electrification and decarbonization of electricity. I examined these effects under multiple revenue redistribution instruments. I show that while carbon taxation robustly accelerates emission reduction, its effectiveness and welfare effects are highly dependent on the tax path and revenue recycling instrument. I also show that welfare effects are unequally distributed and that for a large part this is caused by the electrification margin.

I show that the electrification margin causes the welfare burden to be larger at the lower-end of the income distribution. When subsidy recycling is used, these effects are salient, while they erode when households receive all tax revenues back in the form of a uniform lump-sum transfer. However, this comes at a cost of a lower GDP in the final steady state, caused by a higher maximum tax rate. This effect can however be attenuated by front-loading the tax in combination with lump-sum redistribution. Under this scenario, front-loading does not decrease aggregate welfare, is even more progressive and has a lower GDP decrease than the baseline case.

Overall, the electrification margin for households created distributional effects via heterogeneous carbon intensity of energy demand, that are significant. When policy-makers are intent on reducing emissions, they can best balance this objective with an equity objective through the use of lump-sum redistribution rather than targeted subsidies, and phase the tax in quickly to diminish future costs of the carbon tax. These results underscore that the design of revenue recycling determines who wins and who loses from climate policy. By quantifying the relative importance of the underlying inequality channels, the paper offers guidance for policymakers seeking to balance equity and decarbonization efficiency objectives along the green transition.

References

- Ameli, N., & Brandt, N. (2015). Determinants of households' investment in energy efficiency and renewables: Evidence from the oecd survey on household environmental behaviour and attitudes. *Environmental Research Letters*, 10(4), 044015.
- Ascari, G., Colciago, A., Haber, T., & Wöhrmüller, S. (2025). *Inequality along the european green transition* (Tech. Rep.).
- Benmir, G., & Roman, J. (2022). *The distributional costs of net-zero: A hank perspective* (Tech. Rep.). Working paper.
- Berg, T., Nielsson, U., & Streitz, D. (2024). Too poor to be green? the effects of wealth on the residential heating transformation. *The effects of wealth on the residential heating transformation (May 29, 2024)*.
- Boehl, G., Budianto, F., & Takáts, E. (2024). *The macroeconomics of green transitions* (Tech. Rep.). Bank for International Settlements.

- Boppart, T. (2014). Structural change and the kaldor facts in a growth model with relative price effects and non-gorman preferences. *Econometrica*, 82(6), 2167–2196.
- Coenen, G., Lozej, M., & Priftis, R. (2024). Macroeconomic effects of carbon transition policies: an assessment based on the ecb’s new area-wide model with a disaggregated energy sector. *European Economic Review*, 167, 104798.
- Crucitti, F., Mersch, M., Michaelides, A., & Ostrovnaya, A. (2024). Green technology adoption over the life cycle. *Available at SSRN 5019631*.
- De Nardi, M. (2004). Wealth inequality and intergenerational links. *The Review of Economic Studies*, 71(3), 743–768.
- Douenne, T., Dyrda, S., Hummel, A. J., & Pedroni, M. (2025). *Optimal climate policy with incomplete markets* (Tech. Rep.).
- Dröes, M. I., & Van Der Straten, Y. (2024). Home improvement, wealth inequality, and the energy-efficiency paradox.
- Druedahl, J. (2021). A guide on solving non-convex consumption-saving models. *Computational Economics*, 58(3), 747–775.
- Fridstrøm, L., & Østli, V. (2021). Direct and cross price elasticities of demand for gasoline, diesel, hybrid and battery electric cars: the case of norway. *European Transport Research Review*, 13(1), 3.
- Fried, S. (2026). A macro study of the unequal effects of climate change. *The Economic Journal*, ueaf135.
- Fried, S., Novan, K., & Peterman, W. B. (2018). The distributional effects of a carbon tax on current and future generations. *Review of Economic Dynamics*, 30, 30–46.
- Fried, S., Novan, K., & Peterman, W. B. (2022). Climate policy transition risk and the macroeconomy. *European Economic Review*, 147, 104174.
- Fried, S., Novan, K., & Peterman, W. B. (2024). Understanding the inequality and welfare impacts of carbon tax policies. *Journal of the Association of Environmental and Resource Economists*, 11(S1), S231–S260.
- Gilmore, E. A., & Lave, L. B. (2013). Comparing resale prices and total cost of ownership for gasoline, hybrid and diesel passenger cars and trucks. *Transport Policy*, 27, 200–208.
- Hassler, J., Krusell, P., & Olovsson, C. (2021). Directed technical change as a response to natural resource scarcity. *Journal of Political Economy*, 129(11), 3039–3072.
- Hochmuth, P., Krusell, P., & Mitman, K. (2025). Distributional consequences of becoming climate-neutral.

- Känzig, D. R. (2023). *The unequal economic consequences of carbon pricing* (Tech. Rep.). National Bureau of Economic Research.
- Kuhn, M., & Schlattmann, L. (2024). Distributional consequences of climate policies. *Distributional Consequences of Climate Policies*.
- Labrousse, C., & Perdureau, Y. (2024). Geography versus income: the heterogeneous effects of carbon taxation.
- Levi, S. (2021). Why hate carbon taxes? machine learning evidence on the roles of personal responsibility, trust, revenue recycling, and other factors across 23 european countries. *Energy Research & Social Science*, 73, 101883.
- Olovsson, C., & Vestin, D. (2023). *Greenflation?* (Tech. Rep.). Sveriges Riksbank Working Paper Series.
- Papageorgiou, C., Saam, M., & Schulte, P. (2017). Substitution between clean and dirty energy inputs: A macroeconomic perspective. *Review of Economics and Statistics*, 99(2), 281–290.
- Schlattmann, L. (2024). *Spatial redistribution of carbon taxes* (Tech. Rep.). ECONtribute Discussion Paper.
- Schlöter, L. (2022). Empirical analysis of the depreciation of electric vehicles compared to gasoline vehicles. *Transport Policy*, 126, 268–279.
- Wöhrmüller, S. (2024). Carbon taxation and precautionary savings. *Available at SSRN 4808442*.

Appendix

A Heatpump uptake

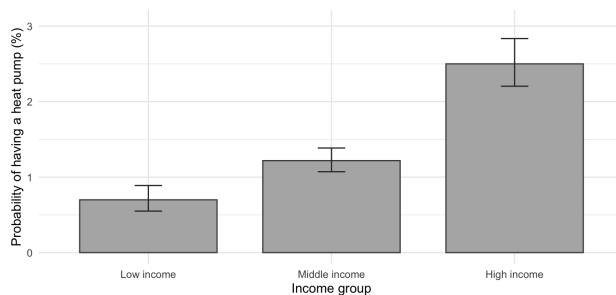


Figure 15: The predicted share of households for different income groups with a heatpump in 2018 using the Woon-dataset from Statistics Netherlands, for homeowners (37750 obs.), mean is 1.41% (dashed line). The correlation persists when controlling for square meters of the house and household age. The correlation is also found for 2021 and 2024.

B Calibration data

DNB household survey

The DNB household survey consists of longitudinal data following about 2000 households. The survey contains multiple topics, such as work, pensions, mortgages, income, assets, liabilities and others. I use information about income, assets and liabilities from the AGI (Aggregated data on income) and AGW (Aggregated data on assets, liabilities and mortgages) datasets. These datasets contain information about for example total assets or income, as a sum of individual components that were asked in the surveys. The data I use is from the years 2010 to 2020.

To estimate the income process, I constrain the sample to the ages 25-65. I delete observations with a negative net labor income or net labor income that exceeds 300.000 euros per year, to not include outliers. For the wealth moments, I treat negative wealth levels as zero-wealth in order to match the models no unsecured borrowing constraint. Net worth is calculated as the sum of checking account, savings, bond holdings, stock holdings and mutual funds, and cars. As I am interested in homeowners and car owners making the investment into a cleaner option, for the wealth moments I only use data on homeowners, but exclude housing wealth from the net worth sum. If I were to include renters, the sum of wealth would be lower and I would underestimate the financial ability of homeowners to make a green investment.

C Solving the household problem

The value function is given by

$$V_j(a, \phi, m, y; P, S) = \max\{V_j^{NoAdjust}(a, \phi, m, y; P, S), V_j^{Electrify}(a, \phi, m, y; P, S) + \mu_{Electrify}\} \quad (39)$$

The preference shock μ_{Adj} causes that this extensive margin becomes a switching probability in the expectation, i.e. before realization of the shock. The probability is given by

$$P^{Electrify}(a, \phi, m, y; P, S) = 1 - \exp\left(-\exp\left(\frac{V_j^{Electrify}(a, \phi, m, y; P, S) + \mu_{Electrify} - V_j^{NoAdjust}(a, \phi, m, y; P, S)}{\sigma_{adj}}\right)\right) \quad (40)$$

I individually solve for the intermediate functions, which I can do for the last living period given the bequest value. Then I use backward induction to solve for the remainder of the periods.

I start by solving the non-adjuster problem. I first solve for the static problem analytically. The dynamic problem then becomes

$$V_j^{NA}(a, \phi, m, y; P, S) = \max_{a'} \tilde{u}(\tilde{c}) + \beta \mathbb{E}[V_{j+1}(a', \phi', m', y'; P', S')] \quad (41)$$

subject to

$$\tilde{w} = wy + (1+r)a + \mathcal{T} - \delta_\phi P^{d_\phi} - \pi(j, t)m - a' \quad (42)$$

$$e = (1-\gamma)\frac{\tilde{w}}{p_\phi} + \gamma\frac{\bar{s}}{A_{s,\phi}} \quad (43)$$

$$c = \gamma\left(\tilde{w} - \frac{p_\phi}{A_{s,\phi}}\bar{s}\right) \quad (44)$$

$$\tilde{c} = c^\gamma (A_{s,\phi}e - \bar{s})^{1-\gamma} \quad (45)$$

$$\tilde{u}(\tilde{c}) = \frac{\tilde{c}^{1-\sigma}}{1-\sigma} \quad (46)$$

$$m' = (1+r+r^m - \pi(j, t))m, \quad \phi' = \phi \quad (47)$$

$$a' \geq 0 \quad (48)$$

where $A_{s,\phi} \equiv \phi A_{s,G} + (1-\phi)A_{s,B}$ and $p_\phi \equiv \phi P^E + (1-\phi)(P^{FF} + \tau_c)$. I solve the non-adjuster problem using the endogenous grid method with upper envelope as described in Druedahl (2021).

Given value function $V_j^{NA}(a, \phi, m, y; P, S)$ and policy functions $a'_j(a, \phi, m, y; P, S)$, $c_j(a, \phi, m, y; P, S)$, $e_j(a, \phi, m, y; P, S)$, I continue by solving for the electrifier value function, by recognizing that this is an interpolation of the non-adjuster problem

$$V_j^{Electrify}(a, \phi, m, y; P, S) = \max_{m'} V_j^{NA}(\tilde{a}, \phi' = 1, m', y; P, S) \quad (49)$$

$$\tilde{a} = a - (P^G(1 - s_{dG}) - \varphi P^B) + m' \quad (50)$$

$$m' \leq \lambda(1 - s_{dG})P^G \quad (51)$$

Which gives policy function $\phi'_j(a, \phi, m, y; P, S), m'_j(a, \phi, m, y; P, S)$ and value function $V_j^{Electrify}(a, \phi, m, y; P, S)$

D Solving the Production side

There are three production sectors. One for the final output good, one for the two durable goods and one for electricity.

D.1 Output-good-producing firm

A representative firm produces output goods which can be used for production or converted in durable goods. Inputs are capital, labor and electricity. The technology is a constant returns to scale CES

$$Y^C(K_C, L_C, E_C, FF_C) = \left[(1 - \psi) (A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c})^{\frac{\nu-1}{\nu}} + \psi A_{MC} M^{\frac{\nu-1}{\nu}} \right]^{\frac{\nu}{\nu-1}} \quad (52)$$

Gives first order conditions

$$r + \delta = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1 - \psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} \alpha_c K_C^{\alpha_c-1} L_C^{1-\alpha_c} \quad (53)$$

$$w = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1 - \psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} (1 - \alpha_c) K_C^{\alpha_c} L_C^{-\alpha_c} \quad (54)$$

$$P^E = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1} \right) \quad (55)$$

$$P^{FF} + \tau_c = \left[(1 - \psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1 - \kappa) FF_C^{\frac{\iota-1}{\iota}-1} \right) \quad (56)$$

where

$$B = A_{KC} K_C^{\alpha_c} L_C^{1-\alpha_c} \quad (57)$$

$$M = \left(\kappa E_C^{\frac{\iota-1}{\iota}} + (1 - \kappa) FF_C^{\frac{\iota-1}{\iota}} \right)^{\frac{\iota}{\iota-1}}. \quad (58)$$

D.2 Durable good producers

In this economy, the durable goods can be produced by converting the consumption good (with cost 1, as its the numeraire) at some rate A_G, A_B , such that production functions are given as

$$Y^{DG}(C_G) = A_G C_G \quad (59)$$

$$Y^{DB}(C_B) = A_B C_B \quad (60)$$

These imply input demand functions

$$C_G = \frac{1}{A_G} Y^{DG} \quad (61)$$

$$C_B = \frac{1}{A_B} Y^{DB} \quad (62)$$

and prices $P^G = \frac{1}{A_G}$, $P^B = \frac{1}{A_B}$

D.3 Electricity firm

The profits of the electricity firm are given as follows

$$\begin{aligned} \Pi(K_{E_R}, K_{E_F}, FF_E) = & P_E \left(\alpha_R^{\frac{1}{\lambda}} (A_R K_R)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} \left(A_F [\zeta K_F^{\frac{\omega-1}{\omega}} + (1-\zeta) FF^{\frac{\omega-1}{\omega}}] \right)^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} \\ & - (r + \delta) K_{E_R} - (r + \delta) K_{E_F} - (P^{FF} + \tau_c) FF_E \end{aligned} \quad (63)$$

Equivalently, I can write the problem as follows

$$\Pi = P_E \left(\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} - P_{E_R} E_R - P_{E_F} E_F \quad (64)$$

subject to the technology constraints

$$E_R = A_R K_R \quad (65)$$

$$E_F = A_F [\zeta^{\frac{1}{\omega}} (A_{KF} K_F)^{\frac{\omega-1}{\omega}} + (1-\zeta)^{\frac{1}{\omega}} FF^{\frac{\omega-1}{\omega}}]^{\frac{\omega}{\omega-1}} \quad (66)$$

Which implies I can split up the problem in two parts. First the problem where I minimize cost by choosing K_{E_R} (K_{E_R}, FF_E) subject to the constraint that production needs to equal some amount E_R (E_F). This problem gives me input demand functions for K_{E_R} (K_{E_R}, FF_E) in terms of E_R (E_F) as well as a price for E_R (E_F), namely P_{E_R} (P_{E_F}).

With these prices I can solve the problem in equation 64, to get the input demand functions for E_F and E_R in terms of some output level Y^E . Combining these with the input functions derived from the other two problems, I get input demand functions: $K_{E_R}(Y^E, r, P^F F, \tau_c)$, $K_{E_F}(Y^E, r, P^F F, \tau_c)$, $FF_E(Y^E, r, P^F F, \tau_c)$, as functions of total output and input prices.

The first step to solving the firm problem for the supply function, is to choose E_F, E_R to minimize cost subject to the constraint of producing some level of electricity $\bar{E}$. The Lagrangian reads

$$\mathcal{L} = P_{E_R} E_R + P_{E_F} E_F + \Lambda (\bar{E} - (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}})$$

The first order conditions are

$$\frac{d\mathcal{L}}{dE_R} \implies \Lambda \alpha_R^{\frac{1}{\lambda}} (E_R)^{\frac{\lambda-1}{\lambda}-1} (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}-1} = P_{E_R} \quad (67)$$

$$\frac{d\mathcal{L}}{dE_F} \implies \Lambda \alpha_F^{\frac{1}{\lambda}} (E_F)^{\frac{\lambda-1}{\lambda}-1} (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}-1} = P_{E_F} \quad (68)$$

$$\frac{d\mathcal{L}}{d\Lambda} \implies \bar{E} = (\alpha_R^{\frac{1}{\lambda}} E_R^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}})^{\frac{\lambda}{\lambda-1}} \quad (69)$$

Dividing through

$$\frac{P_{E_R}}{P_{E_F}} = \frac{\alpha_R^{\frac{1}{\lambda}} (E_R)^{\frac{\lambda-1}{\lambda}-1}}{\alpha_F^{\frac{1}{\lambda}} (E_F)^{\frac{\lambda-1}{\lambda}-1}} \quad (70)$$

Solve for E_R and plug in

$$\left(\frac{P_{E_R}}{P_{E_F}}\right)^{-\lambda} \frac{\alpha_R}{\alpha_F} E_F = E_R \implies \bar{E} = \left(\alpha_R^{\frac{1}{\lambda}} \left(\left(\frac{P_{E_R}}{P_{E_F}}\right)^{-\lambda} \frac{\alpha_R}{\alpha_F} E_F\right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}} E_F^{\frac{\lambda-1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}$$

Rewriting:

$$\bar{E} = \left(\alpha_R^{\frac{1}{\lambda}} \left(\left(\frac{P_{E_R}}{P_{E_F}}\right)^{-\lambda} \frac{\alpha_R}{\alpha_F}\right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}} E_F \quad (71)$$

such that

$$E_F = \frac{\bar{E}}{\left(\alpha_R^{\frac{1}{\lambda}} \left(\left(\frac{P_{E_R}}{P_{E_F}}\right)^{-\lambda} \frac{\alpha_R}{\alpha_F}\right)^{\frac{\lambda-1}{\lambda}} + \alpha_F^{\frac{1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}}} \quad (72)$$

where input demand scales linearly in the amount produced $\bar{E}$, taking all prices as given.

This implies that for any output $\bar{E}$, input demand is given by

$$E_F = \alpha_F \left(\frac{P_{E_F}}{P_E}\right)^{-\lambda} \bar{E} \quad (73)$$

$$E_R = \alpha_R \left(\frac{P_{E_R}}{P_E}\right)^{-\lambda} \bar{E} \quad (74)$$

and the natural price index for E is

$$P_E = (\alpha_R P_{E_R}^{1-\lambda} + \alpha_F P_{E_F}^{1-\lambda})^{\frac{1}{1-\lambda}} \quad (75)$$

Similarly, for the intermediate electricity from renewable and fossil, from the first order conditions, prices for these are given by marginal costs

$$P_{E_R} = \frac{r + \delta}{A_R} \quad (76)$$

$$P_{E_F} = \frac{1}{A_F} (\zeta((r + \delta)/A_{KF})^{1-\omega} + (1 - \zeta)(P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \quad (77)$$

and input demands are

$$K_R = \frac{1}{A_R} E_R \quad (78)$$

$$K_F = \zeta \left(\frac{r + \delta}{A_{KF} P_{E_F}}\right)^{-\omega} E_F \quad (79)$$

$$FF_E = (1 - \zeta) \left(\frac{P_{FF} + \tau_c}{P_{E_F}}\right)^{-\omega} E_F \quad (80)$$

E Equilibrium algorithm

E.1 No tax, stationary equilibrium

Given exogenous fossil fuel price P^{FF}

1. Initialization.

- (a) Guess an interest rate r
- (b) Compute the implied price of electricity and the prices of durables as marginal costs:

$$P^E = \left[\alpha_R \left(\frac{r}{A_R} \right)^{1-\lambda} + \alpha_F \left(\frac{1}{A_F} (\zeta r^{1-\omega} + (1-\zeta)(P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \right)^{1-\lambda} \right]^{\frac{1}{1-\lambda}} \quad (81)$$

$$P^G = \frac{1}{A_G}, \quad P^B = \frac{1}{A_B} \quad (82)$$

2. Firm Problem.

- (a) Using a non-linear solver, solve for (K_C, E_C, FF_C, w) such that the labor market clears and the firm's first-order conditions are satisfied:

$$r + \delta = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1-\psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} \alpha_c K_C^{\alpha_c-1} L_C^{1-\alpha_c} \quad (83)$$

$$w = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} (1-\psi) B^{\frac{\nu-1}{\nu}-1} A_{KC} (1-\alpha_c) K_C^{\alpha_c} L_C^{-\alpha_c} \quad (84)$$

$$P^E = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} \kappa E_C^{\frac{\iota-1}{\iota}-1} \right) \quad (85)$$

$$P^{FF} + \tau_c = \left[(1-\psi) B^{\frac{\nu-1}{\nu}} + \psi M^{\frac{\nu-1}{\nu}} \right]^{\frac{1}{\nu-1}} \psi M^{\frac{\nu-1}{\nu}-1} \left(M^{\frac{1}{\iota}} (1-\kappa) FF_C^{\frac{\iota-1}{\iota}-1} \right) \quad (86)$$

3. Household Problem.

- (a) Given the price vector $P = (P^{FF}, P^G, P^B, P^E, r, w)$, compute the value,

$$V_j^{NA}(a, \phi, m, y; P, S), V_j^A(a, \phi, m, y; P, S),$$

and policy functions,

$$\phi'_j(a, \phi, m, y; P, S), m'_j(a, \phi, m, y; P, S), a'_j(a, \phi, m, y; P, S), c_j(a, \phi, m, y; P, S), e_j(a, \phi, m, y; P, S).$$

- (b) Compute the stationary distribution μ such that the bequest condition holds:

$$\int a'(\mathbf{x}) d\mu_J = \int \frac{a}{1+r}, d\mu-1 \quad (87)$$

4. Aggregation.

(a) Compute household aggregates:

$$A_{hh} = \sum \int^{J-1} a(\mathbf{x}), d\mu_j, \quad C_{hh} = \sum \int^{J-1} c, d\mu_j, \quad (88)$$

$$E_{hh} = \sum \int^{J-1} \phi' e, d\mu_j, \quad FF_{hh} = \sum \int^{J-1} (1 - \phi') e, d\mu_j \quad (89)$$

(b) Compute net durable demands:

$$D_{hh}^G = \sum_{j=0}^{J-1} \int [\phi \delta_G + (\phi' - \phi)], d\mu_j \quad (90)$$

$$D_{hh}^B = \sum_{j=0}^{J-1} \int [(1 - \phi) \delta_B - \varphi(\phi' - \phi)], d\mu_j \quad (91)$$

5. Sectoral Output and Input Demands.

(a) Use market clearing for electricity:

$$Y^E = E_{hh} + E_C \quad (92)$$

(b) Compute input demands for electricity production:

$$K_{E_R} = \frac{1}{A_R} \alpha_R \left(\frac{P^{E_R}}{P^E} \right)^{-\lambda} Y^E, \quad (93)$$

$$K_{E_F} = \zeta \left(\frac{r}{P^{E_F}} \right)^{-\omega} \alpha_F \left(\frac{P^{E_F}}{P^E} \right)^{-\lambda} Y^E \quad (94)$$

$$FF_E = (1 - \zeta) \left(\frac{P^{FF} + \tau_c}{P^{E_F}} \right)^{-\omega} \alpha_F \left(\frac{P^{E_F}}{P^E} \right)^{-\lambda} Y^E \quad (95)$$

(c) Prices of intermediate inputs:

$$P_{E_R} = \frac{r}{A_R}, \quad P_{E_F} = \frac{1}{A_F} (\zeta^\omega r^{1-\omega} + (1 - \zeta)^\omega (P_{FF} + \tau_c)^{1-\omega})^{\frac{1}{1-\omega}} \quad (96)$$

6. Durable Goods Sectors.

(a) Use market clearing for durables:

$$Y^{D_G} = D_{hh}^G, \quad Y^{D_B} = D_{hh}^B \quad (97)$$

(b) Compute input demands:

$$C_G = \frac{Y^{D_G}}{A_G}, \quad C_B = \frac{Y^{D_B}}{A_B} \quad (98)$$

7. Equilibrium Conditions.

- (a) Check capital market clearing:

$$A = K_C + K_{E_R} + K_{E_F} \quad (99)$$

- (b) Check aggregate resource constraint:

$$Y = C_{hh} + C_B + C_G + I_{E_R} + I_{E_F} + I_C + P^{FF}(FF_{hh} + F_E) + \int \phi f(j)M - (\phi' - \phi)M', d\mu \quad (100)$$

If either condition fails, update r and return to Step 1(b).

E.2 Transition

1. Initialization.

- (a) Fix a time horizon $t = 0, \dots, T$, tolerance levels $\varepsilon_r > 0$, and a damping parameter $\eta_r \in (0, 1)$.
- (b) Choose an initial guess for the interest-rate path $\{r_t^{(i=0)}\}_{t=0}^T$.
- (c) Specify path for productivity gains $\{A_t\}_{t=0}^T$ and taxes $\{\tau_{ct}\}_{t=0}^T$, and calculate the implied sequences for $\{P_t^E\}_{t=0}^T$ and $\{w_t\}_{t=0}^T$ from the firm problem.

2. Outer iteration $i = 0, 1, 2, \dots$ (until convergence):

- (a) Compute the value function of the stationary equilibrium in the final period T , $V_j^T(a, \phi; Y, P, \mathcal{T})$. By backward induction, taking the price sequences as given, compute the value function $V_j^t(a, \phi; Y, P, \mathcal{T})$ for $t = T - 1, \dots, 0$. Using the policy functions, simulate the distribution forward for T periods starting from the initial distribution μ_0 , being the initial cross-sectional distribution of households μ_0 as the steady state at $t = 0$.
- (b) From the distribution, for each time period t calculate aggregate capital holdings $\int a(r_i)d\mu_t$ and aggregate electricity demand from households $E_{hh} = \int \phi ed\mu_t$.
- (c) **Inner iteration**, $j = 0, 1, 2, \dots$ (until convergence): solve the static production problem to clear the market. For each time period t :
 - i. Guess an interest rate r_j , calculate $P_j^E(r_j)$.
 - ii. Given r_j , $P_j^E(r_j)$ calculate K_C and E_C from the first order conditions of the final consumption good firm.
 - iii. Given E_{hh} , E_C and r_j calculate K^{E_R} and K_{E_F} .
 - iv. Check if $\int a(r_i)d\mu_t = K_C(r_j) + K^{E_R}(r_j) + K_{E_F}(r_j)$. If not, update r_j and go back to ii.

(d) Check if $\{r_t^{(i)}\}_{t=0}^T = \{r_t^{(j)}\}_{t=0}^T$, if not update

$$\{r_t^{(i+1)}\}_{t=0}^T = \eta_r \{r_t^{(i)}\}_{t=0}^T + (1 - \eta_r) \{r_t^{(j)}\}_{t=0}^T$$

and continue with step 1.c.

F Deriving the aggregate resource constraint

Given production functions for durable goods, consumption and electricity

$$D_G : Y^{d_G}(C_G) = A_G C_G \quad (101)$$

$$D_B : Y^{d_B}(C_B) = A_B C_B \quad (102)$$

$$C : Y^C(K_C, L_C, E_C, FF_C) = F^C(K_C, L_C, E_C, FF_C) \quad (103)$$

$$E : Y^E(K_{E_F}, K_{E_R}, FF_E) = F^E(K_{E_F}, K_{E_R}, FF_E) \quad (104)$$

and the fact that fossil fuels are imported, the following markets need to clear²²:

$$\text{Capital:} \quad \sum_{j=1}^{J-1} \int a_t d\mu_j = K_{C,t} + K_{E_R,t} + K_{E_F,t} \quad (105)$$

$$\text{Labor:} \quad \sum_{j=1}^{J-1} \int y d\mu_j = L_C (= \bar{L}) \quad (106)$$

$$\text{Green Durable:} \quad Y^{D_G} = \sum_{j=1}^{J-1} \int [(\phi' - \phi) + \delta_G \phi] d\mu_j \quad (107)$$

$$\text{Brown Durable:} \quad Y^{D_B} = \sum_{j=1}^{J-1} \int [\delta_B(1 - \phi) - \varphi(\phi' - \phi)] d\mu_j \quad (108)$$

$$\text{Fossil Fuel:} \quad Y^{FF} = FF_{HH} + FF_E + FF_C \quad (109)$$

$$\text{Electricity:} \quad Y^E = \sum_{j=1}^{J-1} \int e\phi' d\mu_j + E_C \quad (110)$$

the government budget constraint needs to hold

$$G + \mathcal{T} + s_{ER} K_{ER}(r + \delta) + s_{dG} P^G \int (\phi' - \phi) d\mu = \tau_c [FF_{HH} + FF_E + FF_C] \quad (111)$$

such that the aggregate resource constraint can be derived. Without loss of generality, I here only derive the case for lump-sum redistribution.

²²The conditions read as follows. Total saving must equal total capital demand. Total labor supply must equal labor demand. Production of the green durable must be equal to newly bought green durables plus depreciation (bequests don't go on the market but are passed through as the durable itself), production of brown durable equals depreciation minus what comes on the market (can be negative?). Total fossil fuel equals household usage and fossil fuels used in electricity production, lastly electricity production is equal to household demand plus final consumption good electricity input demand

Starting from the household budget constraint (note that households with $\phi = 1$ are not allowed choose to switch back after electrification, such that the budget constraint can be written as):

$$wy + (1+r)a + \mathcal{T} = c + a' + \phi' e P^E + (1 - \phi') e (P^{FF} + \tau_c) + \phi \delta_G P^{d_G} + (1 - \phi) \delta_B P^{d_B} \quad (112)$$

$$+ (\phi' - \phi)(P^{d_G} - \varphi P^{d_B}) + \underbrace{\phi \pi(j)m - (\phi' - \phi)m'}_{=m(\phi', \phi)}$$

where $(\phi' - \phi)(P^{d_G} - \varphi P^{d_B})$ is the cost of electrification. Furthermore $m(\phi', \phi)$ is the loan installment payment in case of staying (having previously electrified) or receiving the loan in case of electrifying.

In the first step I integrate both sides of the budget constraint

$$wL_C + \int (1+r)a d\mu + \mathcal{T} = C_{hh} + \int a' d\mu + \int \phi' e P^E d\mu + \int (1 - \phi') e (P^{FF} + \tau_c) d\mu + \int \phi \delta_G P^{d_G} d\mu + \int (1 - \phi) \delta_B P^{d_B} d\mu \quad (113)$$

$$+ \int (\phi' - \phi)(P^{d_G} - \varphi P^{d_B}) d\mu + \int m(\phi', \phi) d\mu$$

Starting with the the two durable goods, in the budget constraint, rearrange terms to get these:

$$P^{d_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] \quad \text{and} \quad P^{d_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] \quad (114)$$

which are exactly as in market clearing conditions (7) and (8), such that I can write

$$P^{d_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] = P^{d_B} Y^{d_B} \quad \text{and} \quad P^{d_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] = P^{d_G} Y^{d_G} \quad (115)$$

Furthermore, plugging the production functions for $Y^{d_B} = A_G C_G$ and $Y^{d_G} = A_G C_G$, and noting that in this linear technology, and with price for C equal to one, I get $P^{d_G} = \frac{1}{A_G}$ and $P^{d_B} = \frac{1}{A_B}$ these expressions simply become

$$P^{d_B} \left[\int (1 - \phi) \delta_B - \varphi(\phi' - \phi) d\mu \right] = C_B \quad \text{and} \quad P^{d_G} \left[\int \phi \delta_G + (\phi' - \phi) d\mu \right] = C_G \quad (116)$$

which I can substitute in equation 113 to get

$$wL_C + \int (1+r)a d\mu + \mathcal{T} = C_{hh} + C_B + C_G + \int a' d\mu + \int \phi' e P^E d\mu + \int (1 - \phi') e (P^{FF} + \tau_c) d\mu \quad (117)$$

$$+ \int m(\phi', \phi) d\mu$$

Next, I rewrite the market clearing condition for electricity,

$$\sum_{j=1}^{J-1} \int e \phi' d\mu_j = Y^E - E_C \quad (118)$$

and substitute the RHS in the new aggregate budget constraint. I also take $(1+r)a$ to the right hand side, to get

$$wL_C + \mathcal{T} = C_{hh} + C_B + C_G + \int [a' - (1+r)a]d\mu + (Y^E - E_C)P^E + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \quad (119)$$

Using the capital market clearing condition, I rewrite $\int [a' - (1+r)a]d\mu$ as follows:

$$[a' - (1+r)a]d\mu = K_{t+1} - (1+r)K_t \quad (120)$$

Noting that $K_t = K_{C,t} + K_{E_R,t} + K_{E_F,t}$, I split up aggregate capital to get:

$$K_{t+1} - (1+r)K_t = K_{C,t+1} - K_{C,t} + K_{E_R,t+1} - K_{E_R,t} + K_{E_F,t+1} - K_{E_F,t} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} \quad (121)$$

$$= I_{C,t} + I_{E_R,t} + I_{E_F,t} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} \quad (122)$$

which I substitute in the aggregate budget constraint

$$wL_C + \mathcal{T} = C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{C,t} - rK_{E_R,t} - rK_{E_F,t} + (Y^E - E_C)P^E + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \quad (123)$$

Take $-rK_{C,t}$ and $-P^E E_C$ to the other side to get

$$wL_C + rK_{C,t} + P^E E_C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} - rK_{E_R} - rK_{E_F} + Y^E P^E + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \quad (124)$$

and since standard FOC for the final production good imply that marginal products are equalized with input prices, Eulers theorem implies that $Y^C = wL_C + (\delta + r)K_{C,t} + P^E E_C + (P^{FF} + \tau_c)FF_C$:

$$Y^C + \mathcal{T} = (P^{FF} + \tau_c)FF_C + C_{hh} + C_B + C_G + I_C + I_{E_R} + I_{E_F} - rK_{E_R} - rK_{E_F} + Y^E P^E + \delta K_C + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \quad (125)$$

There is a zero-profit condition in the market for electricity which reads that

$$Y^E P^E - (\delta + r)K_{E_R,t} - (\delta + r)K_{E_F,t} - FF_E(P^{FF} + \tau_c) = 0 \quad (126)$$

So I can substitute for $Y^E P^E - rK_{E_R,t} - rK_{E_F,t}$ in the aggregate budget constraint

$$Y^C + \mathcal{T} = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + (P^{FF} + \tau_c)FF_C + FF_E(P^{FF} + \tau_c) + \delta(K_C + K_{E_R} + K_{E_F}) + \int (1 - \phi')e(P^{FF} + \tau_c)d\mu + \int m(\phi', \phi)d\mu \quad (127)$$

Next I impose the government budget constraint such that $\mathcal{T}$ cancels with $\tau_c \int e(1 - \phi')d\mu + \tau_c FF_E + \tau_c FF_C$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF} FF_C + FF_E P^{FF} + \delta(K_C + K_{ER} + K_{EF}) + P^{FF} \int (1 - \phi')ed\mu + \int m(\phi', \phi)d\mu \quad (128)$$

Impose market clearing for fossil fuels (with abroad) $FF = FF_C + FF_E + \int (1 - \phi')ed\mu$

$$Y^C = C_{hh} + C_B + C_G + I_{C,t} + I_{E_R,t} + I_{E_F,t} + P^{FF} FF + \int m(\phi', \phi)d\mu + \delta(K_C + K_{ER} + K_{EF}) \quad (129)$$

Substitute in

$$\int m(\phi', \phi)d\mu = \int \phi f(j)M - (\phi' - \phi)M'd\mu \quad (130)$$

And rewrite capital investment

$$Y_t^C = C_{hh,t} + C_{B,t} + C_{G,t} + K_{t+1} - K_t + P^{FF} FF_t + \int \phi f(j)M_t - (\phi' - \phi)M_{t+1}]d\mu \quad (131)$$

Which reads as follows. Total production can either be used for consumption of the household directly, can be converted in brown or durable goods, can be invested in capital, can be spent on fossil fuels or spend on repaying the loan. In steady state,

$$Y^C = C_{hh} + C_B + C_G + \delta K + P^{FF} FF + M(r + r_m) \quad (132)$$

G Climate Model

G.1 The Climate Model

The climate module is exactly taken from Douenne et al. (2025). The temperature is given as

$$Z_{t+1} = Z_t + \epsilon(\zeta \mathcal{E}_t - Z_t), \quad (133)$$

where cumulative emissions evolve according to

$$\mathcal{E}_{t+1} = \mathcal{E}_t + E_t^M + E_t^{\text{ex}}. \quad (134)$$

I set Z_0 , ϵ , ζ , $\mathcal{E}_0$, and the exogenous emissions path $\{E_t^{\text{ex}}\}_{t \geq 0}$ externally, and take all values from Douenne et al. (2025).

G.2 Mapping model outcomes to emissions

In the model, I explicitly track fossil fuel demand, FF_t , as the sum of direct household and firm fossil fuel demand and indirect fossil fuel demand in electricity production:

$$FF_t = FF_{E,t} + FF_{C,t} + FF_{HH,t}. \quad (135)$$

The exogenous price of fossil fuels is calibrated as the price of 1000 kWh of energy, where 67% stems from gasoline and 33% from natural gas. The 670 kWh gasoline component corresponds to

$$\frac{670}{8.9} = 75.2 \text{ liters of gasoline,} \quad (136)$$

and the 330 kWh natural gas component corresponds to

$$\frac{330}{10} = 33 \text{ m}^3 \text{ of natural gas.} \quad (137)$$

Using emission factors of 2.3 kg CO₂ per liter of gasoline and 1.9 kg CO₂ per m³ of natural gas, one unit of fossil fuel in the model produces

$$75.2 \cdot 2.3 + 33 \cdot 1.9 = 235.66 \text{ kg CO}_2. \quad (138)$$

Hence, model emissions implied by fossil fuel demand are given by

$$E_t^M = 235.66 \cdot FF_t, \quad (139)$$

with E_t^M measured in kg CO₂.

G.3 Temperature outcomes

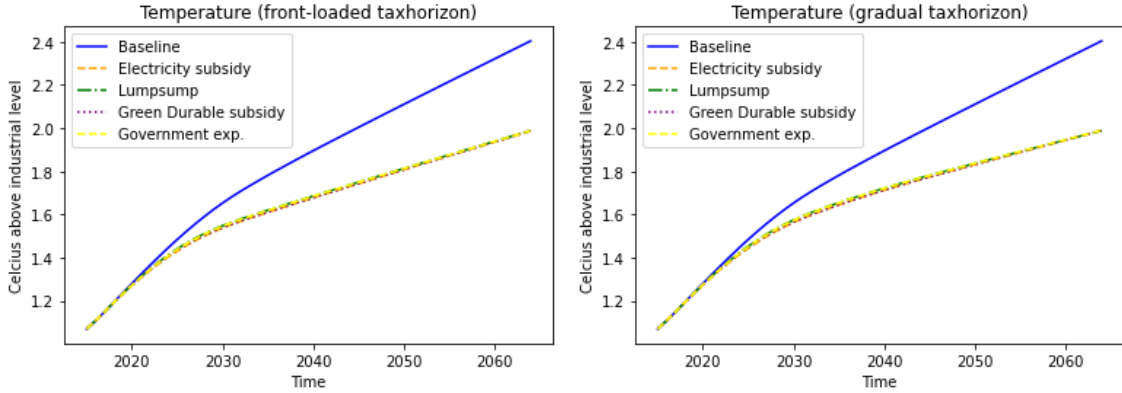


Figure 16: Temperature paths by implementation horizon and revenue-recycling instrument in order to meet the exogenous temperature target (2 degrees celcius)

H Lifetime Consumption Decomposition

I derive the decomposition of lifetime consumption. In doing so, I express the total expected lifetime consumption (plus savings bequests) of a household as the sum of its lifetime income sources, net of energy and durable expenditures.

Step 1: Per-period budget constraint. At age j and time t , the household's budget constraint is:

$$c_{j,t} + P^{\phi_{j,t}} e_{j,t} + a'_{j,t} = \mathcal{T}_t + w_t y_j + (1 + r_t) a_{j,t} - \delta_{\phi_{j,t}} - f(j) m_{j,t} + (\phi'_{j,t} - \phi_{j,t}) [m'_{j,t} - P_t^G + \varphi P^B] \quad (140)$$

where the left-hand side contains consumption, energy expenditure, and savings, and the right-hand side contains transfers, labor income, asset income, maintenance costs, loan repayments, and the net cost of switching to a green durable. To keep notation compact, define:

$$EE_{j,t} \equiv P^{\phi_{j,t}} e_{j,t} \quad (141)$$

$$DE_{j,t} \equiv \delta_{\phi_{j,t}} + f(j) m_{j,t} - (\phi'_{j,t} - \phi_{j,t}) [m'_{j,t} - P_t^G + \varphi P^B] \quad (142)$$

as energy expenditure and net durable expenditure respectively. The budget constraint then simplifies to:

$$c_{j,t} = \mathcal{T}_t + w_t y_j + (1 + r_t) a_{j,t} - a'_{j,t} - EE_{j,t} - DE_{j,t} \quad (143)$$

Step 2: Summing over the remaining lifetime. Consider a household of age j_0 at the time the policy is introduced, $t = 0$. This household has $S \equiv J - 1 - j_0$ remaining periods of life. Summing equation (143) over $s = 0, 1, \dots, S$, where time $t = s$ and the household's age at time s is $j_0 + s$:

$$E_0 \sum_{s=0}^S c_{j_0+s,s} = E_0 \sum_{s=0}^S [\mathcal{T}_s + w_s y_{j_0+s} + (1 + r_s) a_{j_0+s,s} - a'_{j_0+s,s} - EE_{j_0+s,s} - DE_{j_0+s,s}] \quad (144)$$

Step 3: Telescoping the asset terms. Using the fact that $a'_{j,t} = a_{j+1,t+1}$, the asset terms form a telescoping sum. Writing them out explicitly:

$$\begin{aligned} E_0 \sum_{s=0}^S [(1 + r_s) a_{j_0+s,s} - a'_{j_0+s,s}] &= E_0 \left[[(1 + r_0) a_{j_0,0} - a_{j_0+1,1}] \right. \\ &\quad + [(1 + r_1) a_{j_0+1,1} - a_{j_0+2,2}] \\ &\quad + \dots \\ &\quad \left. + [(1 + r_S) a_{j_0+S,S} - a_{j_0+S+1,S+1}] \right] \end{aligned}$$

For each $s \geq 1$, the asset $a_{j_0+s,s}$ appears as $-a_{j_0+s,s}$ from the previous row and $+(1 + r_s) a_{j_0+s,s}$ from its own row, netting to $+r_s \cdot a_{j_0+s,s}$. The final term $a_{j_0+S+1,S+1} = a_{J,T}$ is the stock of assets left at death, which I denote as the bequest a_J . The telescoping sum therefore equals:

$$E_0 \sum_{s=0}^S [(1 + r_s) a_{j_0+s,s} - a'_{j_0+s,s}] = (1 + r_0) a_{j_0,0} + E_0 \left[\sum_{s=1}^S r_s a_{j_0+s,s} - a_J \right] \quad (145)$$

Step 4: Lifetime consumption identity. Substituting (145) back into (144) and moving the bequest a_J to the left-hand side:

$$E_0 \underbrace{\sum_{s=0}^S c_{j_0+s, s} + a_J}_{\text{Lifetime Consumption}} = E_0 \left[\underbrace{(1+r_0)a_{j_0,0}}_{\text{initial wealth}} + \underbrace{\sum_{s=0}^S w_s y_{j_0+s}}_{\text{labor income}} + \underbrace{\sum_{s=1}^S r_s a_{j_0+s, s}}_{\text{asset income}} + \underbrace{\sum_{s=0}^S \mathcal{T}_s}_{\text{transfers}} - \underbrace{\sum_{s=0}^S EE_{j_0+s, s}}_{\text{energy exp.}} - \underbrace{\sum_{s=0}^S DE_{j_0+s, s}}_{\text{durable exp.}} \right] \quad (146)$$

This identity states that total lifetime consumption (including bequests) equals initial wealth plus the present value of all income flows, net of energy and durable expenditures. Initial wealth $(1+r_0)a_{j_0,0}$ is identical across the policy and no-policy case at $t=0$, since the policy is introduced unexpectedly.

Step 5: Change in lifetime consumption. Taking the difference $\Delta(\cdot)$ between the policy scenario and the no-policy baseline, and noting that initial wealth cancels:

$$\begin{aligned} \Delta E_0 \left[\sum_{s=0}^S c_{j_0+s, s} + a_J \right] &= \Delta \sum_{s=0}^S \mathcal{T}_s + \Delta E_0 \sum_{s=0}^S w_s y_{j_0+s} + \Delta E_0 \sum_{s=1}^S r_s a_{j_0+s, s} \\ &\quad - \Delta E_0 \sum_{s=0}^S EE_{j_0+s, s} - \Delta E_0 \sum_{s=0}^S DE_{j_0+s, s} \end{aligned} \quad (147)$$

which we write compactly as:

$\Delta \text{LC} = \Delta \text{Transfers} + \Delta \text{Labor income} + \Delta \text{Asset income} - \Delta \text{Energy expenditure} - \Delta \text{Durable expenditure}$

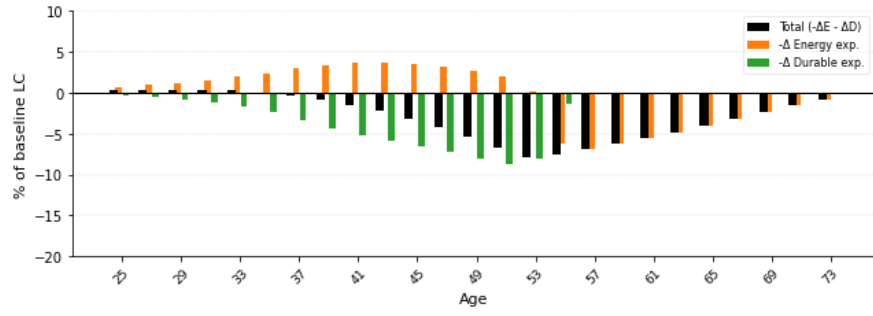
(148)

where ΔLC denotes the change in lifetime consumption relative to the baseline. A positive value of ΔLC means the policy increases the household's total lifetime consumption capacity. This decomposition allows us to isolate the channels through which climate policy affects household welfare: general equilibrium effects on wages and asset returns, direct redistribution through transfers, and expenditure effects arising from the electrification margin.

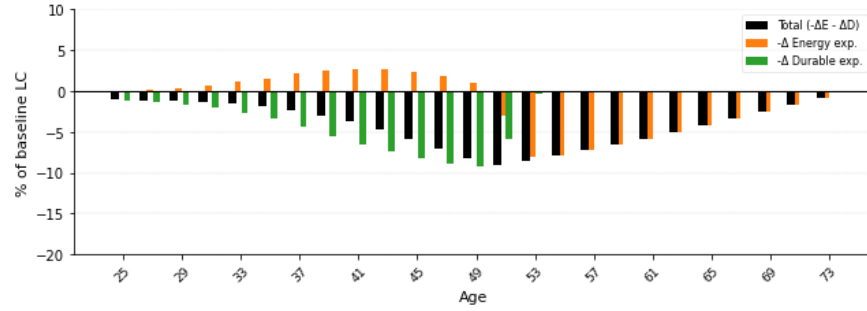
I Extra graphs

I.1 Heterogeneous expenditure changes

Figure 10 for durable subsidy and government expenditure



(a) Durable subsidy



(b) Government expenditure

Figure 17: Heterogeneous expenditure responses across the age space for $z = 0$.

I.2 GDP and prices for combinations of lump-sum and electricity subsidy

First the aggregate outcomes,

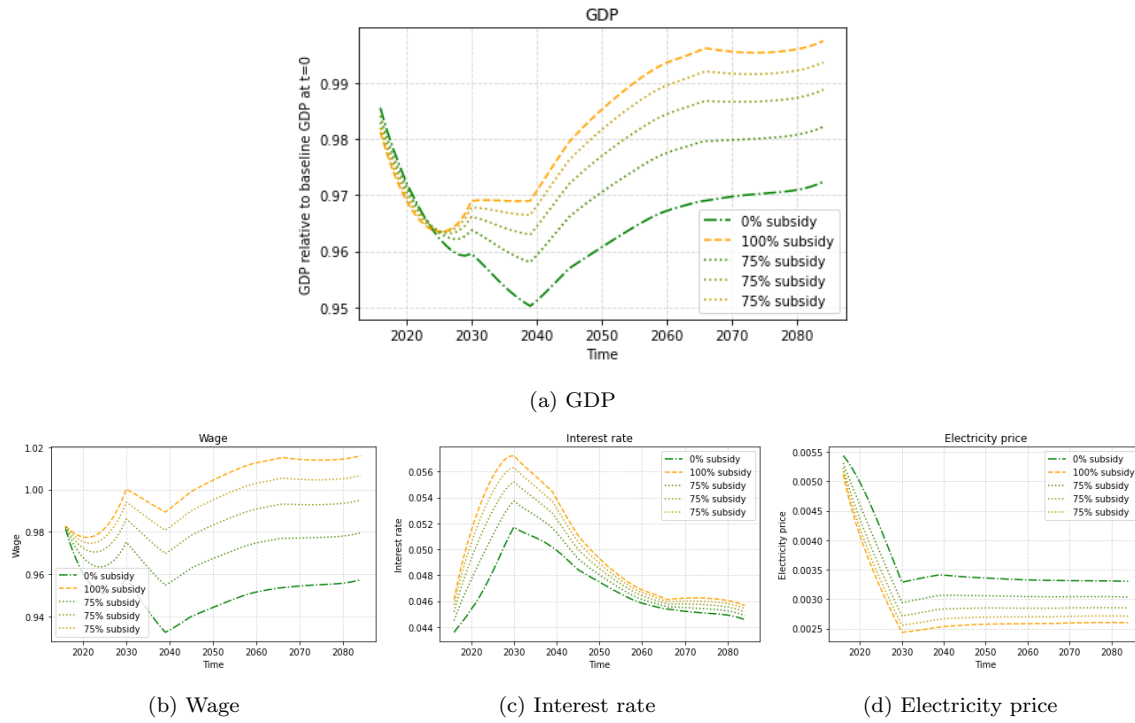


Figure 18: GDP (top) and prices (bottom) over the course of the transition for both subsidies and a combination of the two subsidies

the policies, lump-sum redistribution and tax rate

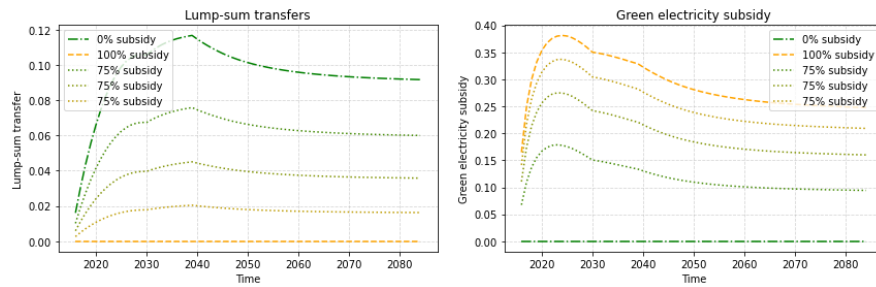


Figure 19: Policies. Right: Lump-sum, left: subsidy

and welfare outcomes

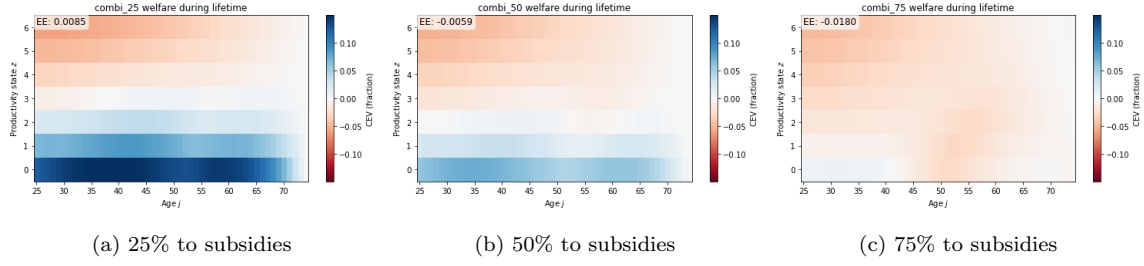


Figure 20: Caption

I.3 GDP and prices for subsidy combinations

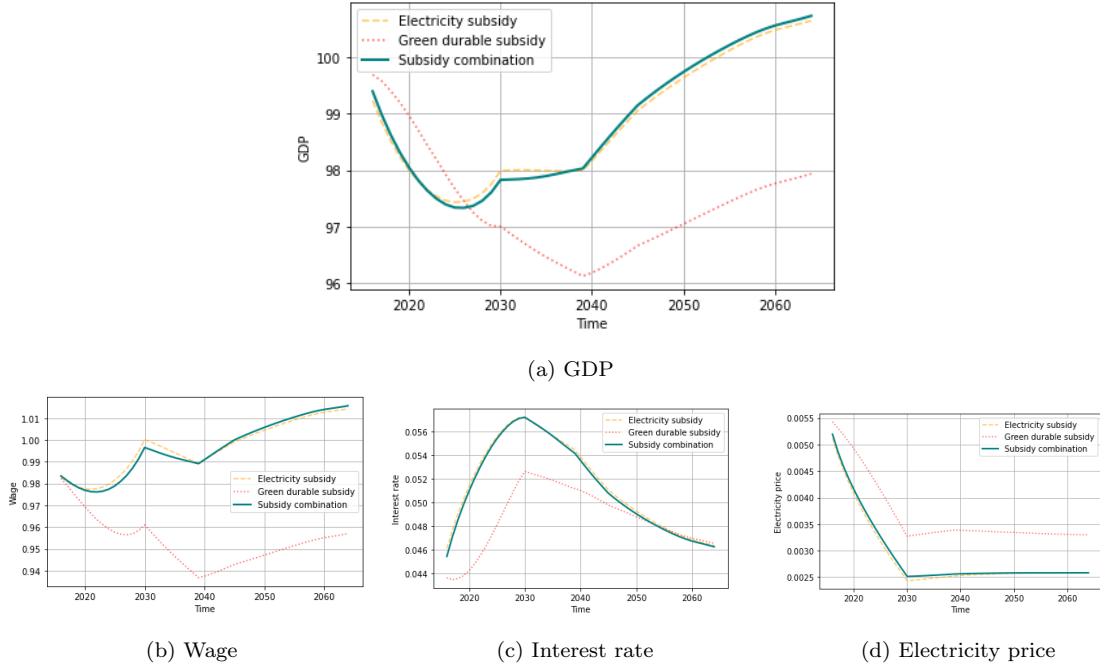


Figure 21: GDP (top) and prices (bottom) over the course of the transition for both subsidies and a combination of the two subsidies

J Robustness

Idiosyncratic income risk A natural source of concern arises from the fact that the lump-sum redistribution scheme alleviates existing distortions, such as idiosyncratic risk and inequalities and is therefore welfare enhancing. In order to test whether the result found above holds true under stricter assumptions,

I solve the model for an economy with the same income states but without any risk, i.e. the transition matrix is now an identity matrix, holding everything else constant and re-analyze the welfare results. I find that both the aggregate as well as the distributional incidence of welfare losses remain qualitatively and quantitatively within the same magnitude under the assumption of no idiosyncratic risk. The aggregate gain in welfare from lump-sum distribution reduces from 2.7% to 2.44%. This still does not speak to the concern that lump-sum might increase aggregate welfare as it alleviates current inequalities, even absent distortions such as risk. However, the conclusion I can draw, and the benefit to lump-sum that is clear compared to all other instruments is that lump-sum is the only instrument which is not regressive. That is, in all cases other than lump-sum the electrification margin causes poor households to suffer the greatest welfare losses as explained above. Lump-sum recycling is the only instrument that alleviates this welfare burden caused by the electrification margin for the poor.

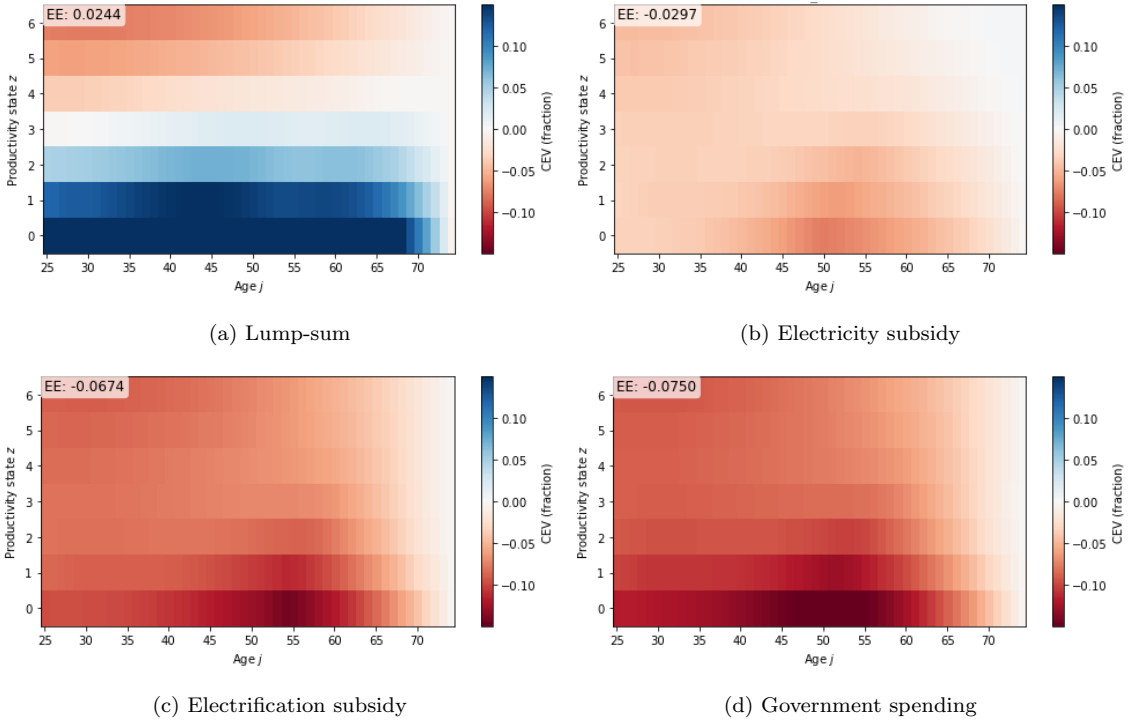


Figure 22: Replication of figure 7 without income risk

EoS energy and non-energy factors, ν A higher elasticity of substitution would imply more substitutability between energy and the capital-labor input in the production of the final consumption good. In this case, an increase in energy prices is expected to have a less severe negative effect on GDP and wages. I reanalyze GDP and wage decreases when adjusting ν from 0.2 to 0.4 but while keeping the carbon tax

constant. This allows for a clean comparison of the robustness of the economic distortions caused by the tax with respect to the EoS between energy and other inputs. Figure 23 shows that results stay the same quantitatively and qualitatively.

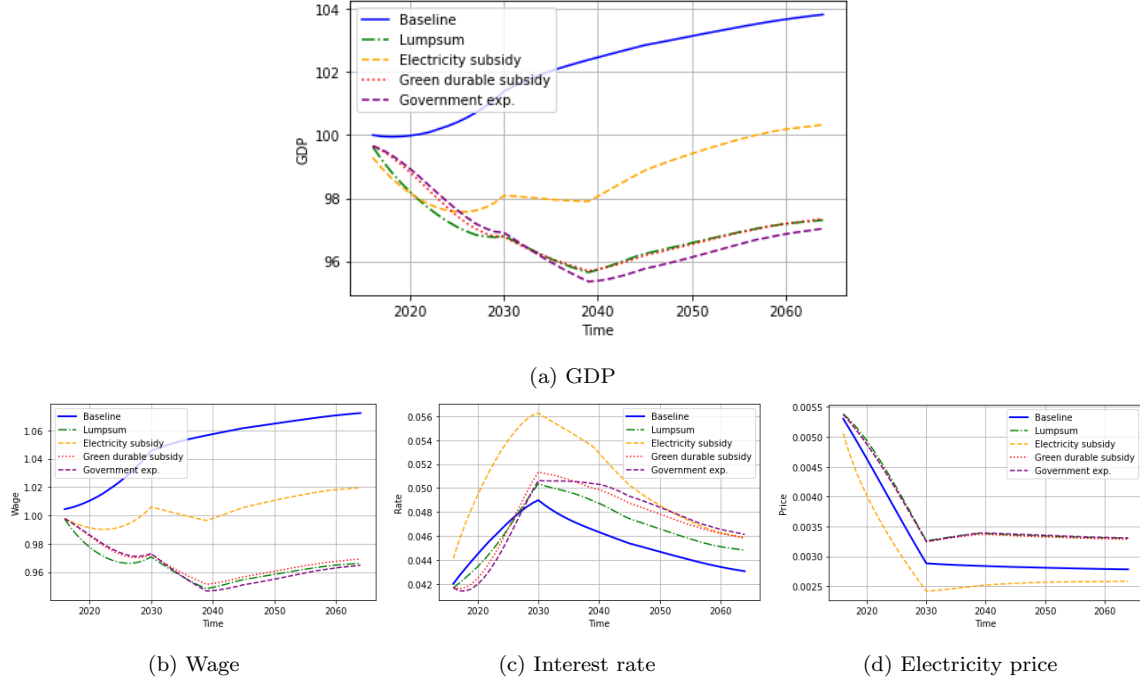


Figure 23: Replication of figure 6 with elasticity of substitution between energy and other inputs, ν , equal to 0.4.

EoS renewable and fossil electricity, λ The elasticity of substitution governs the extent to which the electricity firm can substitute towards renewable sources when the carbon tax rises. A higher λ is therefore expected to lead to a lower peak tax rate, but the degree to which it does could depend on the instrument, i.e. whether or not renewable electricity is subsidized. In order to answer this question, I analyse the cases of a high and low λ , namely 5 and 0.5, in line with ranges used in Hochmuth et al. (2025). In figure 24, I compare tax rates in the baseline regime ($\lambda = 1.8$), with a high and low elasticity regime.

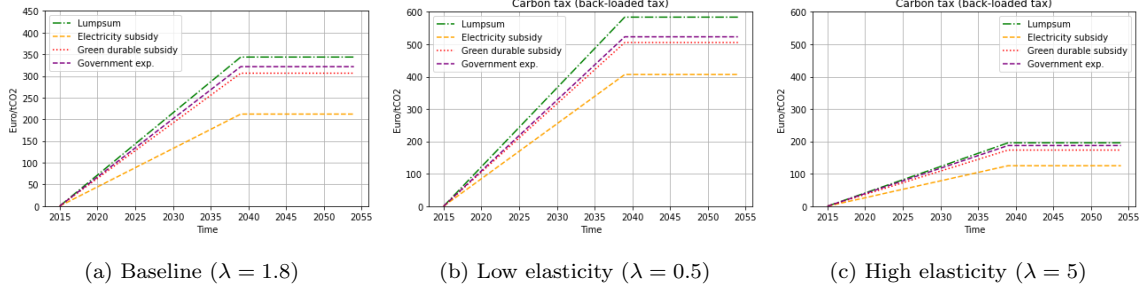


Figure 24: Tax rate under several EoS

Results are in line with findings from Hochmuth et al. (2025), who find that this elasticity of substitution is an important determinant of the endogenous tax rate required to transition to a low-emission economy. Importantly, I do not find any interaction effects with the instruments. That is, the effect of λ on the maximum tax rate is the same for each instrument. This ensures the robustness of assessing between instruments.

Down-payment constraint, λ^{LTV} The salience of the financing frictions is determined by the down payment constraint. In the current calibration, a 10% down-payment is required. I check robustness of the results with alleviating the down-payment constraint, that is if everyone is allowed to borrow against the full amount against the market interest rate, implying a risk premium of 0%.

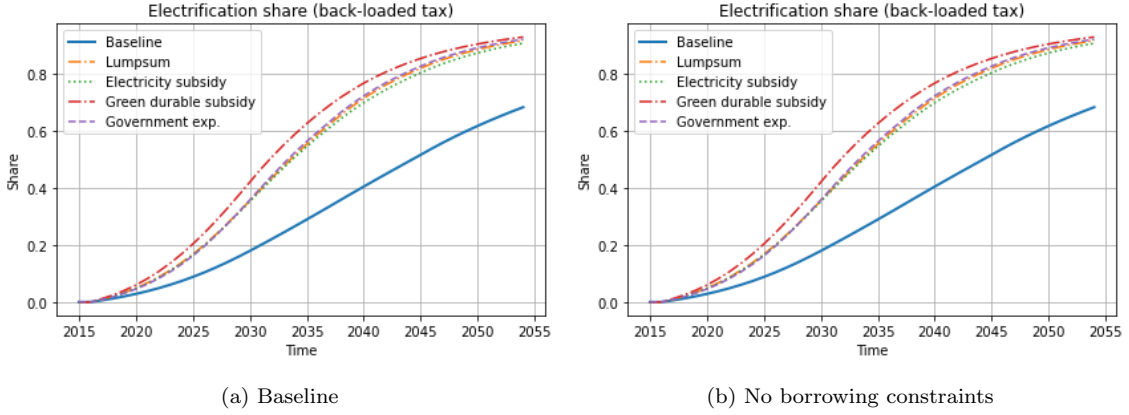


Figure 25: Household electrification

This shows that few households are affected by alleviating constraints. This can be explained by the fact that borrowing is still expensive, as interest rates rise sharply over the transition (figure 6, panel (c)). The high cost of borrowing makes that alleviating risk premia or increasing the space of borrowing does not

affect borrowing at the margin. It does make borrowing for households who are already taking out credit cheaper. This means that the effects for the lower-end of the income distribution who do not electrify are not driven by arbitrary assumptions on the down-payment constraint.